Samkedemia
EDUCATION

ULTIMATE SOLUTION BOOK:
ALL EXERCISES & PAST EXAMS

Research Methods 3

KU LEUVEN • BACHELOR OF BUSINESS ADMINISTRATION

2027 EDITION

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About Me

I'll make sure you pass this exam. I know exactly what it takes to get there.



BACKGROUND

- My name is **Jovan Samardzic**, I am 26 years old
- Got my **MSc in Statistics at KU Leuven**, graduated **Magna Cum Laude**
- Recipient of the **Science@Leuven Scholarship**

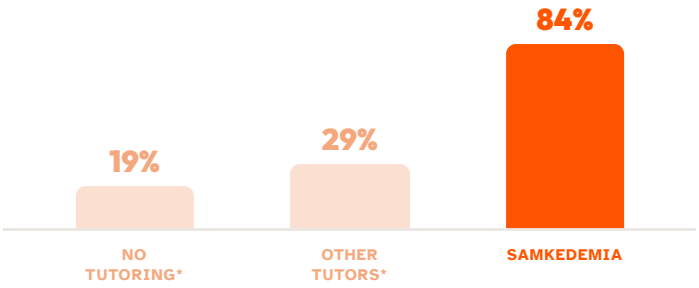
STYLE

- Founder of **Samkedemia**, helping students pass their exams
- Worked with **200+ university students** from all kinds of backgrounds
- **Plain language, not textbook theory** - just what actually makes sense

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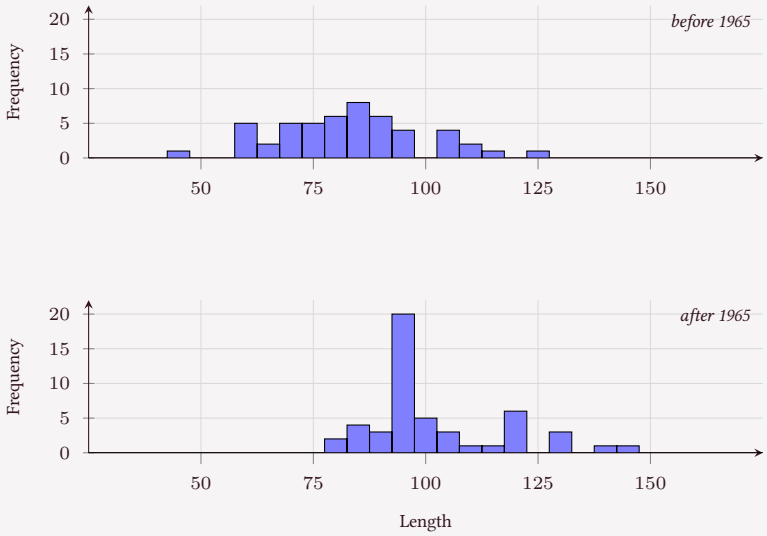
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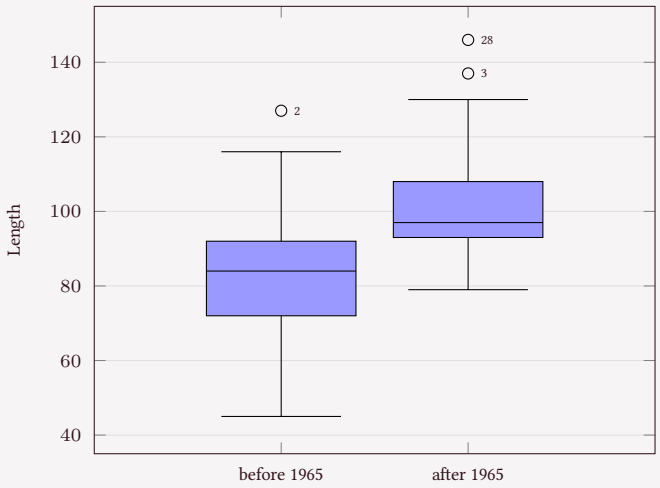
Chapter 1 - Hypothesis Testing

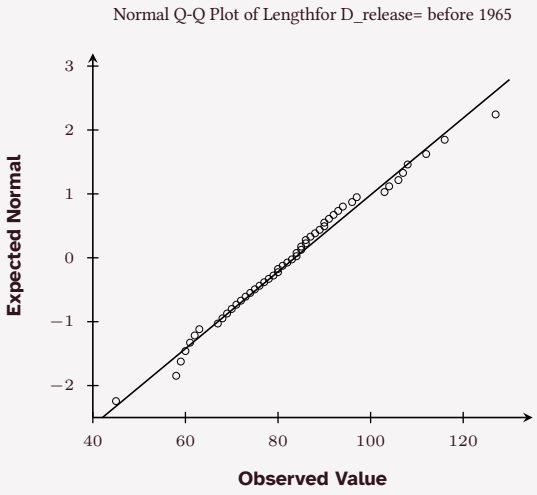
Question 1

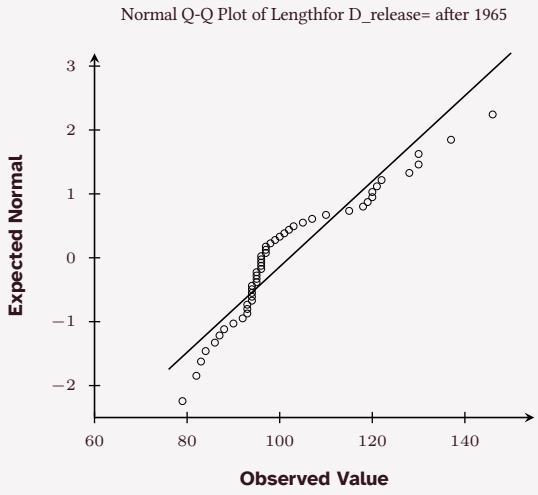
We consider data on a simple random sample of 100 movies (see file **movies.sav**). In a first analysis we want to test whether movies released before 1965 are on average longer than movies released after 1965. The variable **length** measures the length of a movie in minutes. The variable **D_release** indicates whether a movie was released before or after 1965.

Analysis with SPSS yields the following results:









Group Statistics

		N	Mean	Std. Deviation	Std. Error Mean
Length	before 1965	50	83.64	16.624	2.351
	after 1965	50	102.10	14.941	2.113

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means						
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
									Lower	Upper
Length	Equal variances assumed	.351	.555	-5.840	98	.000	-18.460	3.161	-24.733	-12.187
	Equal variances not assumed			-5.840	96.904	.000	-18.460	3.161	-24.734	-12.186

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a. Test whether movies released after 1965 are longer than movies released before 1965.

b. Discuss whether the assumptions of the independent-samples t-test are satisfied.

c. Explain what you can conclude from the "95% confidence interval of the difference" in the output of the independent samples t-test.

d. Indicate which test could be used to test whether the median length of movies released after 1965 is larger than the median length of movies released before 1965. Discuss whether the assumptions of this test are satisfied.

Tip: Independent-Samples t-test

The **independent-samples t-test** is a parametric test used to compare the population means of two independent samples.

$$H_0 : \mu_1 = \mu_2 \qquad H_A : \mu_1 \neq \mu_2$$

It comes with two assumptions:

- The two samples are independent (different subjects in each group, not the same subjects measured twice).
- $n_1, n_2 > 30$, so the CLT ensures the test statistic has an approximate t-distribution in each group. Otherwise, the variable must be Normally distributed within each group for the test statistic to have an exact t-distribution.

This course does not derive the test statistic by hand - it is only ever read off SPSS output.

Tip: Levene's Test

Levene's test checks whether the population variances of two or more groups are equal - it is used to check an assumption of the independent-samples t-test (and of ANOVA), not as a test of interest on its own.

$$H_0 : \sigma_1^2 = \sigma_2^2 \qquad H_A : \sigma_1^2 \neq \sigma_2^2$$

This course only ever reads F and $Sig.$ off SPSS output: if $Sig. \geq \alpha$, we do not reject equal variances; if $Sig. < \alpha$, we reject equal variances.

Solution 1: a. Testing whether movies released after 1965 are longer

We define the two variables as follows:

$$X_1 = \text{length (in minutes) of a movie released before 1965}$$

$$X_2 = \text{length (in minutes) of a movie released after 1965}$$

Step 1: Define the hypotheses.

We test whether the average length is larger for movies released after 1965:

$$H_0 : \mu_1 \geq \mu_2$$

$$H_A : \mu_1 < \mu_2$$

$$\alpha = 0.05$$

Step 2: State the estimator.

We use the independent-samples t-test. Before reading the test value, we first check Levene's test for equality of variances, reported in the same Independent Samples Test table:

$$F = 0.351$$

$$Sig. = 0.555$$

Since $Sig. \geq \alpha$, hence we do not reject equal variances, and read the "Equal variances assumed"

row in the Independent Samples Test table.

Step 3: Find the test value.

As mentioned above, we look at the "Equal variances assumed" row of the Independent Samples Test table. In the columns t and df, we see:

$$t = -5.840$$

$$df = 98$$

Step 4: Find the p-value.

SPSS reports a two-tailed significance in the same row:

$$Sig.\ (\text{2-tailed}) = 0.000$$

The mean difference between group 1 and group 2 is negative, so the test value t is negative as well, which matches the direction that H_A claims. We therefore divide the reported significance by two to obtain the one-tailed p-value:

$$p = \frac{0.000}{2} \approx 0.000$$

Step 5: Make the decision.

Since $p < \alpha$, we reject H_0.

Answer: We conclude with 95% confidence that movies released after 1965 are, on average, longer than movies released before 1965.

Solution 1: b. Discuss whether the assumptions of the test are satisfied

The independent-samples t-test makes two assumptions:

- The two samples are independent.
- $n_1, n_2 > 30$, so the CLT ensures the test statistic has an approximate t-distribution in each group. Otherwise, the variable must be Normally distributed within each group.

Step 1: Check independence.
The before-1965 and after-1965 groups consist of different movies, not the same movies measured twice, so the two samples are independent. This assumption is satisfied.

Step 2: Check Normality/CLT.
Both groups have $n = 50 > 30$, so the CLT ensures the test statistic has an approximate t-distribution regardless of the shape of the population distribution. This assumption is satisfied. The boxplot does show a few outliers in each group (case 2 before 1965; cases 3 and 28 after 1965), but with $n = 50$ in each group the CLT still applies.

Answer: Both assumptions of the independent-samples t-test are satisfied.

Solution 1: c. Interpreting the 95% confidence interval of the difference

We read the 95% confidence interval of the difference from the same "Equal variances assumed" row of the Independent Samples Test table, in the "Lower" and "Upper" columns under 95% Confidence Interval of the Difference:

$$[-24.733,\ -12.187]$$

We are 95% confident that the true population mean length of movies released before 1965 is,

on average, between 24.733 and 12.187 minutes *shorter* than movies released after 1965. Since the entire interval lies below 0, this is consistent with the conclusion in part a: movies released after 1965 are on average longer.

Answer: With 95% confidence, movies released before 1965 are on average between 12.187 and 24.733 minutes shorter than movies released after 1965.

Tip: Mann-Whitney-Wilcoxon Test

The **Mann-Whitney-Wilcoxon test** is a non-parametric test used to compare the population medians of two independent samples, when the data is ordinal or non-normal quantitative.

$$H_0 : \text{median}_1 = \text{median}_2 \qquad H_A : \text{median}_1 \neq \text{median}_2$$

It comes with four assumptions:

- The variable is quantitative or ordinal.
- The two samples are independent.
- The two groups' distributions have the same shape (they can only differ by a shift in location).
- $n_1 + n_2 \geq 20$.

We prefer it over the independent-samples t-test when Normality/CLT fails, or the median is more meaningful than the mean.

Solution 1: d. Testing the median, and its assumptions

Step 1: Choose the test.

Movies released before 1965 and movies released after 1965 are two independent samples of different movies, not the same movies measured twice, and length is a quantitative variable. We are asked about the median, not the mean, so we use the non-parametric Mann-Whitney-Wilcoxon test to compare the two groups' medians, instead of the independent-samples t-test.

Step 2: Check the assumptions.

The Mann-Whitney-Wilcoxon test makes four assumptions:

Step 2.1: Check the variable type.

We are measuring length, which is a quantitative variable, so this assumption is satisfied.

Step 2.2: Check independence.

Movies released before 1965 and movies released after 1965 are different movies, not the same movies measured twice, so the two samples are independent. This assumption is satisfied.

Step 2.3: Check the same-shape assumption.

Looking at the two histograms, both distributions are unimodal and roughly bell-shaped, though the after-1965 group has a taller, narrower peak. Both boxplots also show outliers (case 2 before 1965; cases 3 and 28 after 1965). The two distributions look reasonably similar in shape, so this assumption is roughly satisfied.

Step 2.4: Check the sample-size assumption.

$n_1 + n_2 = 50 + 50 = 100 \geq 20$, so this assumption is satisfied.

Answer: We would use the Mann-Whitney-Wilcoxon test; all of its assumptions are satisfied (or roughly satisfied).

Question 2

The file **movies.sav** also contains the rating (ranging from 1=very bad to 4=very good) of each movie from a certain critic. Some movie directors claim that this critic is unfair in that he assigns a negative rating (=rating lower than or equal to 2.5) to more than 65% of the movies.

The frequency distribution of movie ratings of the critic equals:

Rating

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1.0	8	8.0	8.0	8.0
	1.5	14	14.0	14.0	22.0
	2.0	19	19.0	19.0	41.0
	2.5	27	27.0	27.0	68.0
	3.0	25	25.0	25.0	93.0
	3.5	6	6.0	6.0	99.0
	4.0	1	1.0	1.0	100.0
Total		100	100.0	100.0	

a. Test whether more than 65% of the movies in the population receive a rating from the critic lower than or equal to 2.5.

b. Discuss whether the assumptions of the test are satisfied.

Tip: Binomial Test

The **binomial test** is a non-parametric test used to test a population proportion π, for one category of a two-category variable in a single sample, against a hypothesized value π_0.

$$H_0 : \pi = \pi_0 \qquad H_A : \pi \neq \pi_0$$

It comes with three assumptions:

- Each observation is a Bernoulli trial: a yes/no outcome with the same probability π every

time.

- The number of trials n is fixed in advance.
- The observations are independent of each other.

The estimator we use has an exact binomial distribution under H_0:

$$X \sim \text{Binom}(n,\ \pi_0) = \text{number of observations (out of } n\text{) in the category of interest}$$

Solution 2: a. Testing whether more than 65% of movies receive a negative rating

Step 1: Define the hypotheses.

Let π denote the population proportion of movies that receive a rating of 2.5 or lower from this critic. The movie directors claim this is more than 65%, so we test:

$$H_0 : \pi \leq 0.65$$

$$H_A : \pi > 0.65$$

$$\alpha = 0.05$$

Step 2: State the estimator.

We use the binomial test, with the estimator and its distribution under H_0:

$$X \sim \text{Binom}(n,\ \pi_0) = \text{number of movies (out of } n\text{) that receive a rating of 2.5 or lower}$$

Step 3: Find the test value.

Here $n = 100$ movies were sampled, and $\pi_0 = 0.65$ from H_0. In the Rating table, we read the Cumulative Percent column at the row for a rating of 2.5, which shows that 68.0% of the 100 movies received a rating of 2.5 or lower:

$$x = 68$$

Step 4: Find the p-value.

Since H_A says the proportion is higher than 0.65, we need the probability of a value at least as large as x:

$$p = P(X \geq 68 \mid H_0) = 1 - \texttt{binomcdf}(100,\ 0.65,\ 67) \approx 0.3029$$

Step 5: Make the decision.

Since $p \geq \alpha$, we cannot reject H_0.

Answer: We cannot conclude, with 95% confidence, that more than 65% of the movies in the population receive a rating of 2.5 or lower from this critic - the population proportion could be 65% or lower.

Solution 2: b. Discuss whether the assumptions of the test are satisfied

The binomial test makes three assumptions:

- Each observation is a Bernoulli trial: a yes/no outcome with the same probability π every time.
- The number of trials n is fixed in advance.
- The observations are independent of each other.

Step 1: Check the Bernoulli trials.

Each movie's rating is a yes/no outcome (rating $\leq$ 2.5 or not), with the same probability π across movies, since every movie is rated by the same critic on the same scale. This assumption is satisfied.

Step 2: Check the fixed number of trials.

The number of movies sampled, $n = 100$, was fixed in advance, before observing the ratings.

This assumption is satisfied.

Step 3: Check independence.

The 100 movies in movies.sav are a simple random sample, so the observations are independent of each other. This assumption is satisfied.

Answer: All three assumptions of the binomial test are satisfied.

Question 3

The following cross-table shows for a random sample ($n = 183$) how many males and females would spend more than 500 euro when they buy a new smartphone.

gender * D_budget Crosstabulation

Count

		D_budget		
		less than 500 euro	more than 500 euro	Total
gender	male	46	29	75
	female	57	51	108
Total		103	80	183

a. Test whether the proportion of males who spend more than 500 euro on a new smartphone differs from the proportion of females who spend more than 500 euro.

b. Discuss whether the assumptions of the test are satisfied.

Tip: Chi-Square Test for Proportions

The **chi-square test** is a non-parametric test used when the outcome is categorical (nominal), not quantitative. It is used to test whether the proportions of a categorical outcome differ between two or more independent samples.

$$H_0 : \pi_1 = \pi_2 = \dots \quad \text{(population proportions are equal)}$$

$$H_A : \text{at least one } \pi_i \text{ is different}$$

It comes with two assumptions:

- All expected counts are larger than 1.
- At most 20% of the cells have an expected count smaller than 5.

Tip: Chi-Square Test: Mechanics

In order to do the chi-square test, we use the estimator χ^2. Here are the steps to find it, and its distribution under H_0:

1. Build the table of observed counts O, with the groups as rows and the outcome categories as columns (or vice versa).
2. Compute the expected count for every cell, assuming H_0 is true:
$$E = \frac{\text{row total} \times \text{column total}}{\text{grand total}}$$
3. Calculate the estimator:
$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Under H_0:

$$\chi^2 \sim \chi^2_{(r-1)(k-1)}$$

with r the number of rows and k the number of columns in the table.

Solution 3: a. Testing whether the proportion who spend more than 500 euro differs between males and females

Step 1: Define the hypotheses.

Let $\pi_{more|i}$ be the proportion who would spend more than 500 euro on a new smartphone among people of gender i:

$$\pi_{more|M} = \text{proportion of males who would spend more than 500 euro}$$

$$\pi_{more|F} = \text{proportion of females who would spend more than 500 euro}$$

We test whether this proportion differs between males and females:

$$H_0 : \pi_{more|M} = \pi_{more|F}$$

$$H_A : \pi_{more|M} \neq \pi_{more|F}$$

$$\alpha = 0.05$$

Step 2: State the estimator.

Males and females are two independent groups of people, not the same people measured twice, and whether someone would spend more than 500 euro is a categorical variable (yes/no), not quantitative. This is exactly what the chi-square test for proportions is for: comparing a categorical outcome across two or more independent groups.

For the chi-square test, we use the estimator χ^2:

$$\chi^2 = \sum \frac{(O-E)^2}{E}, \qquad E = \frac{\text{row total} \times \text{column total}}{\text{grand total}}$$

Under H_0, χ^2 has a chi-square distribution with $(r-1)(k-1)$ degrees of freedom.

Step 3: Find the test value.

None of the expected counts are given, so we compute all four of them:

$$E_{less|M} = \frac{75 \times 103}{183} \approx 42.2131 \quad E_{more|M} = \frac{75 \times 80}{183} \approx 32.7869$$

$$E_{less|F} = \frac{108 \times 103}{183} \approx 60.7869 \quad E_{more|F} = \frac{108 \times 80}{183} \approx 47.2131$$

With every expected count now known, we compute χ^2:

$$\chi^2 = \frac{(46-42.2131)^2}{42.2131} + \frac{(29-32.7869)^2}{32.7869} + \frac{(57-60.7869)^2}{60.7869} + \frac{(51-47.2131)^2}{47.2131} \approx 1.3168$$

with $df = (2-1)(2-1) = 1$.

Step 4: Find the p-value.

The chi-square test is always right-tailed, whatever H_A says, since only a large χ^2 value counts as evidence against H_0:

$$p = P(\chi^2 \geq 1.3168) = \chi^2\texttt{cdf}(\texttt{1.3168, 1E99, 1}) \approx 0.2512$$

Step 5: Make the decision.
The p-value is large, $p > 0.05$, so we cannot reject H_0.

Answer: We cannot conclude, with 95% confidence, that the proportion of males who would spend more than 500 euro on a new smartphone differs from the proportion of females - the proportions could be equal.

Solution 3: b. Discuss whether the assumptions of the test are satisfied

The chi-square test makes two assumptions:

- All expected counts are larger than 1.
- At most 20% of the cells have an expected count smaller than 5.

Step 1: Check that all expected counts are larger than 1.
The four expected counts are 42.2131, 32.7869, 60.7869, and 47.2131, all larger than 1. This assumption is satisfied.

Step 2: Check that at most 20% of the cells have an expected count smaller than 5.
None of the four expected counts are smaller than 5 (0% of the cells), so this assumption is satisfied too.

Answer: Both assumptions of the chi-square test are satisfied.

Question 4

The following cross-table shows for a random sample of 130 persons the relation between net household income and how much they would spend when they buy a new smartphone.

net household income * budget Crosstabulation

		budget					
		between 0 and 200 euro	between 200 and 500 euro	between 500 and 700 euro	between 700 and 900 euro	more than 900 euro	Total
less than 1000	Count	3	14	10	8	0	35
	Expected Count	7.8	12.1	6.5	5.7	3.0	35.0
between 1000 and 1500	Count	7	5	2	4	3	21
	Expected Count	4.7	7.3	3.9	3.4	1.8	21.0
between 1500 and 2000	Count	7	12	5	5	3	32
	Expected Count	7.1	11.1	5.9	5.2	2.7	32.0
between 2000 and 3000	Count	10	10	6	2	1	29
	Expected Count	6.5	10.0	5.4	4.7	2.5	29.0
more than 3000	Count	2	4	1	2	4	13
	Expected Count	2.9	4.5	2.4	2.1	1.1	13.0
Total	Count	29	45	24	21	11	130
	Expected Count	29.0	45.0	24.0	21.0	11.0	130.0

a. Test whether the variables **net household income** and **budget** are statistically dependent.

b. Discuss whether the assumptions of the test are satisfied, and indicate how you can proceed when the assumptions of the test are not satisfied.

Tip: Chi-Square Test of Independence

The **chi-square test** is a non-parametric test used to test whether two categorical (nominal) variables are statistically independent.

H_0 : the two variables are independent $\qquad$ H_A : the two variables are dependent

It comes with two assumptions:

- All expected counts are larger than 1.
- At most 20% of the cells have an expected count smaller than 5.

It is mathematically the same test as the chi-square test for proportions: testing whether the proportions of one variable are equal across the categories of the other is the same as testing whether the two variables are independent.

Tip: Chi-Square Test: Mechanics

In order to do the chi-square test, we use the estimator χ^2. Here are the steps to find it, and its distribution under H_0:

1. Build the table of observed counts O, with the groups as rows and the outcome categories as columns (or vice versa).
2. Compute the expected count for every cell, assuming H_0 is true:

$$E = \frac{\text{row total} \times \text{column total}}{\text{grand total}}$$

3. Calculate the estimator:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Under H_0:

$$\chi^2 \sim \chi^2_{(r-1)(k-1)}$$

with r the number of rows and k the number of columns in the table.

Solution 4: a. Testing whether net household income and budget are statistically dependent

Step 1: Define the hypotheses.

We test whether the two categorical variables are related:

$$H_0 : \text{net household income and budget are statistically independent}$$

$$H_A : \text{net household income and budget are statistically dependent}$$

$$\alpha = 0.05$$

Step 2: State the estimator.

Net household income and budget are both categorical variables (income classes and budget classes), and neither is a grouping/treatment variable - the question just asks whether the two are related. This is exactly what the chi-square test of independence is for.

For the chi-square test, we use the estimator χ^2:

$$\chi^2 = \sum \frac{(O-E)^2}{E}, \qquad E = \frac{\text{row total} \times \text{column total}}{\text{grand total}}$$

Under H_0, χ^2 has a chi-square distribution with $(r-1)(k-1)$ degrees of freedom.

Step 3: Find the test value.

The expected counts are already shown in the crosstable above; computing them to more decimal places, e.g. for income less than 1000 and a budget between 0 and 200 euro:

$$E = \frac{35 \times 29}{130} \approx 7.8077$$

Repeating this for every cell gives:

income \ budget	0-200	200-500	500-700	700-900	>900
less than 1000	7.8077	12.1154	6.4615	5.6538	2.9615
between 1000 and 1500	4.6846	7.2692	3.8769	3.3923	1.7769
between 1500 and 2000	7.1385	11.0769	5.9077	5.1692	2.7077
between 2000 and 3000	6.4692	10.0385	5.3538	4.6846	2.4538
more than 3000	2.9000	4.5000	2.4000	2.1000	1.1000

With every expected count now known, we compute χ^2:

$$\chi^2 = \frac{(3-7.8077)^2}{7.8077} + \cdots + \frac{(4-1.1)^2}{1.1} \approx 26.3015$$

with $df = (5-1)(5-1) = 16$.

Step 4: Find the p-value.
The chi-square test is always right-tailed, whatever H_A says, since only a large χ^2 value counts as evidence against H_0:

$$p = P(\chi^2 \geq 26.3015) = \chi^2\texttt{cdf}(\texttt{26.3015, 1E99, 16}) \approx 0.0499$$

Step 5: Make the decision.
The p-value is small, $p < 0.05$, so we reject H_0.

Note that this p-value is very close to 0.05. With the rounded expected counts from the SPSS table, we would instead find:

$$\chi^2 \approx 26.2034 \qquad p \approx 0.0512$$

and we would not reject H_0.

Answer: We conclude with 95% confidence that net household income and budget are statistically dependent.

Solution 4: b. Discuss whether the assumptions of the test are satisfied, and how to proceed

The chi-square test makes two assumptions:

- All expected counts are larger than 1.
- At most 20% of the cells have an expected count smaller than 5.

Step 1: Check that all expected counts are larger than 1.
The smallest expected count is 1.1 (income more than 3000, budget more than 900 euro), which

is larger than 1. This assumption is satisfied.

Step 2: Check that at most 20% of the cells have an expected count smaller than 5.
13 of the 25 expected counts are smaller than 5:

$$\frac{13}{25} = 52\% > 20\%$$

so this assumption is *not* satisfied, and we cannot trust the result of part a.

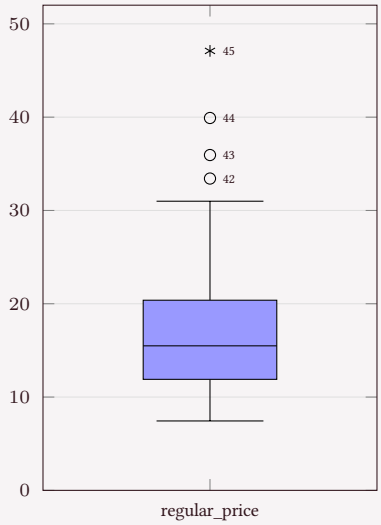
Step 3: How to proceed.
We can merge neighbouring categories, so that each cell contains more observations and the expected counts become larger. For example, we merge the two highest income classes (between 2000 and 3000, and more than 3000) and the two highest budget classes (between 700 and 900 euro, and more than 900 euro). This gives a 4×4 table in which the smallest expected count is 3.88 and only 2 of the 16 cells (12.5%) have an expected count smaller than 5, so both assumptions are satisfied. We then redo the chi-square test on this smaller table.

Answer: The first assumption is satisfied, but the second is not: 52% of the cells have an expected count smaller than 5. We can merge neighbouring income and budget classes until both assumptions are satisfied, and redo the test.

Question 5

The file **price_wine.sav** contains the regular price and the promotion price for a random sample of 45 wines in a specialized wine store.

The following graph shows the distribution of the regular price in the sample:



We also compute descriptive statistics and the frequency distribution for the variable **regular_price**:

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
regular_price	45	7.44	47.10	17.7298	8.97385
Valid N (listwise)	45				

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regular_price

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	7.44	1	2.2	2.2	2.2
	7.81	1	2.2	2.2	4.4
	7.98	1	2.2	2.2	6.7
	8.18	1	2.2	2.2	8.9
	9.30	2	4.4	4.4	13.3
	9.92	1	2.2	2.2	15.6
	10.04	1	2.2	2.2	17.8
	10.29	1	2.2	2.2	20.0
	11.03	1	2.2	2.2	22.2
	11.53	1	2.2	2.2	24.4
	11.90	3	6.7	6.7	31.1
	12.39	2	4.4	4.4	35.6
	13.01	1	2.2	2.2	37.8
	13.14	1	2.2	2.2	40.0
	14.25	1	2.2	2.2	42.2
	14.58	1	2.2	2.2	44.4
	14.87	1	2.2	2.2	46.7
	15.49	2	4.4	4.4	51.1
	16.11	1	2.2	2.2	53.3
	16.36	1	2.2	2.2	55.6
	16.86	1	2.2	2.2	57.8
	17.10	1	2.2	2.2	60.0
	17.35	1	2.2	2.2	62.2
	17.48	1	2.2	2.2	64.4
	17.72	1	2.2	2.2	66.7
	17.97	1	2.2	2.2	68.9
	18.59	1	2.2	2.2	71.1
	20.20	1	2.2	2.2	73.3
	20.38	1	2.2	2.2	75.6
	22.19	1	2.2	2.2	77.8
	22.68	1	2.2	2.2	80.0
	22.93	1	2.2	2.2	82.2
	23.55	1	2.2	2.2	84.4
	28.51	1	2.2	2.2	86.7
	30.37	1	2.2	2.2	88.9
	30.99	1	2.2	2.2	91.1
	33.42	1	2.2	2.2	93.3
	35.94	1	2.2	2.2	95.6
	39.91	1	2.2	2.2	97.8
	47.10	1	2.2	2.2	100.0
Total		45	100.0	100.0	

a. Use a one-sample t-test to test whether the average regular price of wines in the store is larger than 14 euro. Discuss whether the assumptions of the test are satisfied.

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b. Use a sign test to test whether the median price of regular wines in the store is larger than 14 euro. Discuss whether the assumptions of the test are satisfied.

c. Use a signed-rank test to test whether the median price of regular wines in the store is larger than 14 euro. Make the necessary computations in excel or SPSS. Discuss whether the assumptions of the test are satisfied.

Tip: One-Sample t-test

The **one-sample t-test** is a parametric test used to compare the population mean of one sample to a fixed value μ_0.

$$H_0 : \mu = \mu_0 \qquad H_A : \mu \neq \mu_0$$

It comes with two assumptions:

- $n > 30$, so the CLT ensures the test statistic has an approximate t-distribution.
- Otherwise, the variable must be Normally distributed for the test statistic to have an exact t-distribution.

Tip: One-Sample t-test: Mechanics

In order to do the one-sample t-test, we use the estimator T. Here are the steps to find it, and its distribution under H_0:

1. Find the sample mean $\overline{x}$ and the sample standard deviation s of the n observations.
2. Calculate the estimator:

$$T = \frac{\overline{X} - \mu_0}{S/\sqrt{n}}$$

Under H_0:

$$T \sim t_{n-1}$$

Solution 5: a. Testing whether the average regular price is larger than 14 euro (one-sample t-test)

Step 1: Define the hypotheses.

Let μ be the population mean regular price of the wines in the store. We test whether it is larger than 14 euro:

$$H_0 : \mu \leq 14$$

$$H_A : \mu > 14$$

$$\alpha = 0.05$$

Step 2: State the estimator.

For the one-sample t-test, we use the estimator T:

$$T = \frac{\overline{X} - 14}{S/\sqrt{n}}$$

Under H_0, T has a t-distribution:

$$T \sim t_{n-1}$$

This is exact if the variable is Normally distributed, and approximate if $n > 30$ by the CLT.

Step 3: Find the test value.

In the Descriptive Statistics table, we read the sample size, the sample mean and the sample standard deviation of regular_price:

$$n = 45$$

$$\overline{x} = 17.7298$$

$$s = 8.97385$$

Plugging these into the estimator:

$$t = \frac{17.7298 - 14}{8.97385/\sqrt{45}} = \frac{3.7298}{1.3377} \approx 2.7881$$

with $df = n - 1 = 44$.

Step 4: Find the p-value.
Since H_A says the mean is larger, we need the probability of a value at least as large as t:

$$p = P(T \geq 2.7881) = \texttt{tcdf}(2.7881,\ \texttt{1E99},\ 44) \approx 0.0039$$

Step 5: Make the decision.
The p-value is small, $p < 0.05$, so we reject H_0.

Here we also have to check the assumptions of the one-sample t-test:

- $n > 30$, so the CLT ensures the test statistic has an approximate t-distribution.
- Otherwise, the variable must be Normally distributed for the test statistic to have an exact t-distribution.

The boxplot shows that regular_price is right-skewed, with a few outliers at the top, so the variable is not Normally distributed. But here $n = 45 > 30$, so the CLT still ensures that the test statistic has an approximate t-distribution, and the assumptions are satisfied.

Answer: We conclude with 95% confidence that the average regular price of the wines in the store is larger than 14 euro.

Tip: Sign Test

The **sign test** is a non-parametric test used when data is ordinal or quantitative. It is used in two cases:

1. Compare a population median of one sample to a fixed value med_0.

$$H_0 : \text{median} = med_0 \qquad H_A : \text{median} \neq med_0$$

2. Compare the population medians of two paired samples.

$$H_0 : \text{median}_x = \text{median}_y \qquad H_A : \text{median}_x \neq \text{median}_y$$

It comes with one assumption:

- The variable is quantitative or ordinal.

It is less powerful than the signed-rank test, because it only uses the direction of each deviation, not its size. We use it instead of the signed-rank test whenever $n \leq 20$, since the signed-rank test's Normal approximation is not valid below that threshold, while the sign test's Binom distribution is exact for any n.

Tip: Sign Test: Mechanics

In order to do the sign test, we use the estimator B. Here are the steps to find it, and its distribution under H_0:

1. Compute the deviation $x_i - med_0$ for every observation.
2. Drop every observation where that deviation is exactly 0. Let n be the number of observations left.
3. Calculate the estimator:

$$B = \text{number of observations where } x_i - med_0 > 0$$

Under H_0:

$$B \sim \text{Binom}(n,\ 0.5)$$

Solution 5: b. Testing whether the median regular price is larger than 14 euro (sign test)

Step 1: Define the hypotheses.

We test whether the median regular price is larger than 14 euro:

$$H_0 : \text{Median}_{price} \leq 14$$

$$H_A : \text{Median}_{price} > 14$$

$$\alpha = 0.05$$

Step 2: State the estimator.

For the sign test, we use the estimator B:

$$B = \text{number of observations where } price_i - 14 > 0$$

Under H_0, B is exactly Binom:

$$B \sim \text{Binom}(n,\ 0.5)$$

Step 3: Find the test value.

None of the prices equal 14, so all $n = 45$ observations are kept. In the frequency table, the Cumulative Percent column shows that 40.0% of the wines, i.e. 18 of the 45, have a price of 13.14 or lower, and the next price is 14.25. So the other 27 wines have a price larger than 14:

$$b = 45 - 18 = 27$$

Step 4: Find the p-value.

Since H_A says the median is larger, we need the probability of a value at least as large as b:

$$p = P(B \geq 27) = 1 - \texttt{binomcdf}(45,\ 0.5,\ 26) \approx 0.1163$$

Step 5: Make the decision.

The p-value is large, $p > 0.05$, so we cannot reject H_0.

Here we also have to check the assumption of the sign test:

- The variable is quantitative or ordinal.

The regular price is quantitative, so this assumption is satisfied.

Answer: We cannot conclude, with 95% confidence, that the median regular price of the wines in the store is larger than 14 euro - the median could be 14 euro or lower.

Tip: Wilcoxon Signed-Rank Test

The **Wilcoxon signed-rank test** is a non-parametric test used when data is ordinal or non-normal quantitative. It is used in two cases:

1. Compare a population median of one sample to a fixed value med_0.

$$H_0 : \text{median} = med_0 \qquad H_A : \text{median} \neq med_0$$

2. Compare the population medians of two paired samples.

$$H_0 : \text{median}_x = \text{median}_y \qquad H_A : \text{median}_x \neq \text{median}_y$$

It comes with two assumptions:

- The variable is quantitative or ordinal.
- $n > 20$.

We prefer it over the sign test, because it uses the size of each deviation, not just its sign.

Tip: Signed-Rank Test: Mechanics

In order to do the Wilcoxon signed-rank test, we use the estimator B. Here are the steps to find it, and its distribution under H_0:

1. Compute $d_i = |x_i - med_0|$ for every observation.
2. Sort the observations by d_i from smallest to largest, and assign each one a rank equal to its position in that order (the smallest gets rank 1, the next smallest rank 2, and so on).
   2.1 Ties get the average rank (e.g. if the 5th and 6th are tied, both get 5.5).
   2.2 If $d_i = 0$, drop that observation.
3. Calculate the estimator:

$$B = \text{sum of ranks for observations where } x_i - med_0 > 0$$

Under H_0:

$$B \approx \mathcal{N}\left(E(B) = \frac{n(n+1)}{4},\ Var(B) = \frac{n(n+1)(2n+1)}{24}\right)$$

Solution 5: c. Testing whether the median regular price is larger than 14 euro (signed-rank test)

Step 1: Define the hypotheses.

We test whether the median regular price is larger than 14 euro:

$$H_0 : \text{Median}_{price} \leq 14$$

$$H_A : \text{Median}_{price} > 14$$

$$\alpha = 0.05$$

Step 2: State the estimator.

For the Wilcoxon signed-rank test, we use the estimator B:

$$B = \text{sum of ranks for observations where } price_i - 14 > 0$$

Under H_0, B is approximately Normal:

$$B \approx \mathcal{N}\left(E(B) = \frac{n(n+1)}{4},\ Var(B) = \frac{n(n+1)(2n+1)}{24}\right)$$

Step 3: Find the test value.

For each observation, we compute $d_i = |price_i - 14|$ and then rank the d_i from smallest to largest. None of the prices equal 14, so all 45 observations are kept.

Several prices appear more than once (two wines of 15.49, two of 12.39, three of 11.90 and two of 9.30), so their d_i are tied, and each group is given the average of the ranks they would otherwise occupy. Repeating this for every observation gives:

price	$price_i - 14$	$\lvert price_i - 14\rvert$	rank	price	$price_i - 14$	$\lvert price_i - 14\rvert$	rank
14.25	0.25	0.25	1	17.97	3.97	3.97	24
14.58	0.58	0.58	2	9.92	−4.08	4.08	25
13.14	−0.86	0.86	3	18.59	4.59	4.59	26
14.87	0.87	0.87	4	9.30	−4.70	4.70	27.5
13.01	−0.99	0.99	5	9.30	−4.70	4.70	27.5
15.49	1.49	1.49	6.5	8.18	−5.82	5.82	29
15.49	1.49	1.49	6.5	7.98	−6.02	6.02	30
12.39	−1.61	1.61	8.5	7.81	−6.19	6.19	31
12.39	−1.61	1.61	8.5	20.20	6.20	6.20	32
11.90	−2.10	2.10	11	20.38	6.38	6.38	33
11.90	−2.10	2.10	11	7.44	−6.56	6.56	34
11.90	−2.10	2.10	11	22.19	8.19	8.19	35
16.11	2.11	2.11	13	22.68	8.68	8.68	36
16.36	2.36	2.36	14	22.93	8.93	8.93	37
11.53	−2.47	2.47	15	23.55	9.55	9.55	38
16.86	2.86	2.86	16	28.51	14.51	14.51	39
11.03	−2.97	2.97	17	30.37	16.37	16.37	40
17.10	3.10	3.10	18	30.99	16.99	16.99	41
17.35	3.35	3.35	19	33.42	19.42	19.42	42
17.48	3.48	3.48	20	35.94	21.94	21.94	43
10.29	−3.71	3.71	21	39.91	25.91	25.91	44
17.72	3.72	3.72	22	47.10	33.10	33.10	45
10.04	−3.96	3.96	23				

The observed sum of positive ranks:

$$b = 1 + 2 + 4 + 6.5 + 6.5 + 13 + 14 + 16 + 18 + 19 + 20 + 22 + 24 + 26$$

$$+ 32 + 33 + 35 + 36 + 37 + 38 + 39 + 40 + 41 + 42 + 43 + 44 + 45 = 697$$

With $n = 45$:

$$E(B) = \frac{(45)(46)}{4} = 517.5000$$

$$Var(B) = \frac{(45)(46)(91)}{24} = 7848.7500$$

$$SD(B) = \sqrt{7848.7500} = 88.5932$$

Step 4: Find the p-value.

Since H_A says the median is larger, we need the probability of a value at least as large as b:

$$p = P(B \geq 697) = \texttt{normalcdf}(697,\ \texttt{1E99},\ 517.5,\ 88.5932) \approx 0.0214$$

Step 5: Make the decision.

The p-value is small, $p < 0.05$, so we reject H_0.

Here we also have to check the assumptions of the signed-rank test:

- The variable is quantitative or ordinal.
- $n > 20$.

The regular price is quantitative, so the first assumption is satisfied. Here $n = 45 > 20$, so the second assumption is satisfied too.

Answer: We conclude with 95% confidence that the median regular price of the wines in the store is larger than 14 euro.

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Question 6

The file **price_wine.sav** contains both the regular price and the promotion price for a sample of 45 wines from a specialized wine store. The frequency distribution of the regular price minus the promotion price is displayed in the table below:

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regular price minus promotion price

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	0.73	1	2.2	2.2	2.2
	0.86	1	2.2	2.2	4.4
	0.95	1	2.2	2.2	6.7
	1.00	2	4.4	4.4	11.1
	1.03	1	2.2	2.2	13.3
	1.14	1	2.2	2.2	15.6
	1.15	2	4.4	4.4	20.0
	1.18	1	2.2	2.2	22.2
	1.19	1	2.2	2.2	24.4
	1.24	1	2.2	2.2	26.7
	1.30	1	2.2	2.2	28.9
	1.31	1	2.2	2.2	31.1
	1.34	1	2.2	2.2	33.3
	1.36	1	2.2	2.2	35.6
	1.37	1	2.2	2.2	37.8
	1.39	1	2.2	2.2	40.0
	1.40	1	2.2	2.2	42.2
	1.44	1	2.2	2.2	44.4
	1.48	1	2.2	2.2	46.7
	1.50	2	4.4	4.4	51.1
	1.53	1	2.2	2.2	53.3
	1.56	1	2.2	2.2	55.6
	1.59	1	2.2	2.2	57.8
	1.60	1	2.2	2.2	60.0
	1.64	1	2.2	2.2	62.2
	1.64	1	2.2	2.2	64.4
	1.65	1	2.2	2.2	66.7
	1.66	2	4.4	4.4	71.1
	1.68	1	2.2	2.2	73.3
	1.70	1	2.2	2.2	75.6
	1.73	1	2.2	2.2	77.8
	1.81	1	2.2	2.2	80.0
	1.86	1	2.2	2.2	82.2
	1.98	1	2.2	2.2	84.4
	2.39	1	2.2	2.2	86.7
	2.43	1	2.2	2.2	88.9
	2.44	1	2.2	2.2	91.1
	2.49	1	2.2	2.2	93.3
	2.93	1	2.2	2.2	95.6
	3.87	1	2.2	2.2	97.8
	3.94	1	2.2	2.2	100.0
Total		45	100.0	100.0	

Conduct a sign test to test whether the median price reduction in the store for a wine in promo-

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tion is larger than 1 euro. Compute the test statistic of the sign test from the information in the frequency distribution of regular price minus promotion price. Discuss whether the assumptions of the test are satisfied.

Tip: Sign Test

The **sign test** is a non-parametric test used when data is ordinal or quantitative. It is used in two cases:

1. Compare a population median of one sample to a fixed value med_0.

$$H_0 : \text{median} = med_0 \qquad H_A : \text{median} \neq med_0$$

2. Compare the population medians of two paired samples.

$$H_0 : \text{median}_x = \text{median}_y \qquad H_A : \text{median}_x \neq \text{median}_y$$

It comes with one assumption:

- The variable is quantitative or ordinal.

It is less powerful than the signed-rank test, because it only uses the direction of each deviation, not its size. We use it instead of the signed-rank test whenever $n \leq 20$, since the signed-rank test's Normal approximation is not valid below that threshold, while the sign test's Binom distribution is exact for any n.

Tip: Sign Test: Mechanics

In order to do the sign test, we use the estimator B. Here are the steps to find it, and its distribution under H_0:

1. Compute the deviation $x_i - med_0$ for every observation.
2. Drop every observation where that deviation is exactly 0. Let n be the number of observations left.

3. Calculate the estimator:

$$B = \text{number of observations where } x_i - med_0 > 0$$

Under H_0:

$$B \sim \text{Binom}(n,\ 0.5)$$

Solution 6: Testing whether the median price reduction is larger than 1 euro (sign test)

Let X denote the price reduction of a wine in promotion, i.e. the regular price minus the promotion price.

Step 1: Define the hypotheses.
We test whether the median price reduction is larger than 1 euro:

$$H_0 : \text{Median}_X \leq 1$$

$$H_A : \text{Median}_X > 1$$

$$\alpha = 0.05$$

Step 2: State the estimator.
For the sign test, we use the estimator B:

$$B = \text{number of observations where } X_i - 1 > 0$$

Under H_0, B is exactly Binom:

$$B \sim \text{Binom}(n,\ 0.5)$$

Step 3: Find the test value.
In the frequency table, two wines have a price reduction of exactly 1.00, so their deviation from 1 is 0 and they are dropped, leaving $n = 45 - 2 = 43$ observations.

The Cumulative Percent column shows that 11.1% of the wines, i.e. 5 of the 45, have a price reduction of 1.00 or lower. Two of those 5 are exactly 1.00, so 3 wines have a reduction smaller than 1, and all the others have a reduction larger than 1:

$$b = 43 - 3 = 40$$

Step 4: Find the p-value.

Since H_A says the median is larger, we need the probability of a value at least as large as b:

$$p = P(B \geq 40) = 1 - \texttt{binomcdf}(43,\ 0.5,\ 39) \approx 0.0000$$

Step 5: Make the decision.

The p-value is small, $p < 0.05$, so we reject H_0.

Here we also have to check the assumption of the sign test:

- The variable is quantitative or ordinal.

The price reduction is quantitative, so this assumption is satisfied.

Answer: We conclude with 95% confidence that the median price reduction in the store for a wine in promotion is larger than 1 euro.

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Question 7

The file **price_wine.sav** contains both the regular price and the promotion price for a sample of 45 wines from a specialized wine store. Let the variable **X=regular_price-promotion_price** be the price reduction for a wine due to a promotion.

We conduct a signed-rank test in SPSS to test whether the median price reduction in the store for a wine in promotion is larger than 1 euro. The output of the analysis shows that two observations are discarded from the analysis because they have X=1. Applying the signed-rank test to the remaining 43 observations indicates that the sum of negative ranks is equal to 15.5.

a. Compute the sum of positive ranks and use the signed rank test to test whether the median price reduction in the store for a wine in promotion is larger than 1 euro.

b. Discuss whether the assumptions of the test are satisfied.

Tip: Wilcoxon Signed-Rank Test

The **Wilcoxon signed-rank test** is a non-parametric test used when data is ordinal or non-normal quantitative. It is used in two cases:

1. Compare a population median of one sample to a fixed value med_0.

$$H_0 : \text{median} = med_0 \qquad H_A : \text{median} \neq med_0$$

2. Compare the population medians of two paired samples.

$$H_0 : \text{median}_x = \text{median}_y \qquad H_A : \text{median}_x \neq \text{median}_y$$

It comes with two assumptions:

- The variable is quantitative or ordinal.
- $n > 20$.

We prefer it over the sign test, because it uses the size of each deviation, not just its sign.

Tip: Signed-Rank Test: Mechanics

In order to do the Wilcoxon signed-rank test, we use the estimator B. Here are the steps to find it, and its distribution under H_0:

1. Compute $d_i = |x_i - med_0|$ for every observation.
2. Sort the observations by d_i from smallest to largest, and assign each one a rank equal to its position in that order (the smallest gets rank 1, the next smallest rank 2, and so on).
   2.1 Ties get the average rank (e.g. if the 5th and 6th are tied, both get 5.5).
   2.2 If $d_i = 0$, drop that observation.
3. Calculate the estimator:

$$B = \text{sum of ranks for observations where } x_i - med_0 > 0$$

Under H_0:

$$B \approx \mathcal{N}\left(E(B) = \frac{n(n+1)}{4},\ Var(B) = \frac{n(n+1)(2n+1)}{24}\right)$$

Solution 7: a. Testing whether the median price reduction is larger than 1 euro (signed-rank test)

Step 1: Define the hypotheses.

We test whether the median price reduction is larger than 1 euro:

$$H_0 : \text{Median}_X \leq 1$$

$$H_A : \text{Median}_X > 1$$

$$\alpha = 0.05$$

Step 2: State the estimator.

For the Wilcoxon signed-rank test, we use the estimator B:

$$B = \text{sum of ranks for observations where } X_i - 1 > 0$$

Under H_0, B is approximately Normal:

$$B \approx \mathcal{N}\left(E(B) = \frac{n(n+1)}{4},\ Var(B) = \frac{n(n+1)(2n+1)}{24}\right)$$

Step 3: Find the test value.

The two observations with $X = 1$ are dropped, so $n = 43$ observations are ranked. Every observation gets exactly one rank from 1 to 43, so the sum of all ranks is:

$$1 + 2 + \cdots + 43 = \frac{n(n+1)}{2} = \frac{(43)(44)}{2} = 946$$

Each rank belongs either to a positive or to a negative deviation. The sum of negative ranks is 15.5, so the observed sum of positive ranks is:

$$b = 946 - 15.5 = 930.5$$

With $n = 43$:

$$E(B) = \frac{(43)(44)}{4} = 473.0000$$

$$Var(B) = \frac{(43)(44)(87)}{24} = 6858.5000$$

$$SD(B) = \sqrt{6858.5000} = 82.8161$$

Step 4: Find the p-value.

Since H_A says the median is larger, we need the probability of a value at least as large as b:

$$p = P(B \geq 930.5) = \texttt{normalcdf}(\texttt{930.5, 1E99, 473, 82.8161}) \approx 0.0000$$

Step 5: Make the decision.

The p-value is small, $p < 0.05$, so we reject H_0.

Answer: We conclude with 95% confidence that the median price reduction in the store for a wine in promotion is larger than 1 euro.

Solution 7: b. Discuss whether the assumptions of the test are satisfied

The signed-rank test makes two assumptions:

- The variable is quantitative or ordinal.
- $n > 20$.

Step 1: Check the variable type.

The variable X (price reduction) is quantitative, so this assumption is satisfied.

Step 2: Check the sample-size assumption.

Here $n = 43 > 20$, so this assumption is satisfied too.

Answer: Both assumptions of the signed-rank test are satisfied.

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Question 8

20 students take an intensive course in Spanish. To assess the effect of the course, students take a test in Spanish before and after the course.

Let **X** be the test score after taking the course minus the test score before taking the course. The table below shows descriptive statistics for the variable X.

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
score after minus score before course	20	-5.00	7.00	1.4500	3.20321
Valid N (listwise)	20				

The following table shows the frequency distribution of X:

score after minus score before course

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	-5.00	1	5.0	5.0	5.0
	-4.00	1	5.0	5.0	10.0
	-3.00	1	5.0	5.0	15.0
	-2.00	1	5.0	5.0	20.0
	-1.00	2	10.0	10.0	30.0
	1.00	2	10.0	10.0	40.0
	2.00	3	15.0	15.0	55.0
	3.00	4	20.0	20.0	75.0
	4.00	2	10.0	10.0	85.0
	5.00	2	10.0	10.0	95.0
	7.00	1	5.0	5.0	100.0
Total		20	100.0	100.0	

a. Use a parametric test to test whether the average test score of students is higher after the course than before the course. Discuss whether the assumptions of the test are satisfied.

b. Use a sign-test to test whether the median change in the test score (due to taking the course) is positive. Discuss whether the assumptions of the test are satisfied.

c. Use a signed-rank test to test whether the median change in the test score (due to taking

the course) is positive (use SPSS or excel to compute the test statistic). Discuss whether the assumptions of the test are satisfied.

Tip: Paired-Samples t-test

The **paired-samples t-test** is a parametric test used to compare the population means of two paired samples.

$$H_0 : \mu_1 = \mu_2 \qquad H_A : \mu_1 \neq \mu_2$$

It comes with two assumptions:

- $n > 30$, so the CLT ensures the test statistic has an approximate t-distribution.
- Otherwise, the difference between the two paired measurements must be Normally distributed for the test statistic to have an exact t-distribution.

Tip: Paired-Samples t-test: Mechanics

In order to do the paired-samples t-test, we use the estimator T. Here are the steps to find it, and its distribution under H_0:

1. Compute the difference $D_i = x_{1i} - x_{2i}$ for every subject.
2. Find the sample mean $\overline{d}$ and the sample standard deviation s_D of the n differences.
3. Calculate the estimator:

$$T = \frac{\overline{D} - 0}{S_D/\sqrt{n}}$$

Under H_0:

$$T \sim t_{n-1}$$

Solution 8: a. Testing whether the average test score is higher after the course (paired-samples t-test)

Each student takes the test twice, so we define the two paired variables as follows:

$$X_1 = \text{test score of a student after the course}$$

$$X_2 = \text{test score of a student before the course}$$

The variable X in the tables is the difference $D = X_1 - X_2$.

Step 1: Define the hypotheses.

We test whether the average test score is higher after the course than before:

$$H_0 : \mu_1 \leq \mu_2$$

$$H_A : \mu_1 > \mu_2$$

$$\alpha = 0.05$$

Step 2: State the estimator.

For the paired-samples t-test, we use the estimator T:

$$T = \frac{\overline{D} - 0}{S_D/\sqrt{n}}$$

Under H_0, T has a t-distribution:

$$T \sim t_{n-1}$$

This is exact if D is Normally distributed, and approximate if $n > 30$ by the CLT.

Step 3: Find the test value.

In the Descriptive Statistics table, we read the sample size, the sample mean and the sample

standard deviation of the difference X:

$$n = 20$$

$$\overline{d} = 1.4500$$

$$s_D = 3.20321$$

Plugging these into the estimator:

$$t = \frac{1.4500 - 0}{3.20321/\sqrt{20}} = \frac{1.4500}{0.7163} \approx 2.0244$$

with $df = n - 1 = 19$.

Step 4: Find the p-value.
Since H_A says the mean is higher after the course, we need the probability of a value at least as large as t:

$$p = P(T \geq 2.0244) = \texttt{tcdf}(\texttt{2.0244}, \texttt{ 1E99}, \texttt{ 19}) \approx 0.0286$$

Step 5: Make the decision.
The p-value is small, $p < 0.05$, so we reject H_0.

Here we also have to check the assumptions of the paired-samples t-test:

- $n > 30$, so the CLT ensures the test statistic has an approximate t-distribution.
- Otherwise, the difference between the two paired measurements must be Normally distributed for the test statistic to have an exact t-distribution.

Here $n = 20 < 30$, so we cannot rely on the CLT, and the difference X must be Normally distributed. The frequency table shows values between -5 and 7, most of them between 1 and 3, without extreme outliers, so a Normal distribution is plausible. But with only 20 observations and no Normality test, we cannot be sure, so this assumption is doubtful.

Answer: We conclude with 95% confidence that the average test score is higher after the course than before. Since $n = 20$ is small and we cannot check whether the difference is Normally distributed, this conclusion should be treated with caution.

Tip: Sign Test

The **sign test** is a non-parametric test used when data is ordinal or quantitative. It is used in two cases:

1. Compare a population median of one sample to a fixed value med_0.

$$H_0 : \text{median} = med_0 \qquad H_A : \text{median} \neq med_0$$

2. Compare the population medians of two paired samples.

$$H_0 : \text{median}_x = \text{median}_y \qquad H_A : \text{median}_x \neq \text{median}_y$$

It comes with one assumption:

- The variable is quantitative or ordinal.

It is less powerful than the signed-rank test, because it only uses the direction of each deviation, not its size. We use it instead of the signed-rank test whenever $n \leq 20$, since the signed-rank test's Normal approximation is not valid below that threshold, while the sign test's Binom distribution is exact for any n.

Tip: Sign Test: Mechanics

In order to do the sign test, we use the estimator B. Here are the steps to find it, and its distribution under H_0:

1. Compute the deviation $x_i - med_0$ for every observation.
2. Drop every observation where that deviation is exactly 0. Let n be the number of observa-

tions left.

3. Calculate the estimator:

$$B = \text{number of observations where } x_i - med_0 > 0$$

Under H_0:

$$B \sim \text{Binom}(n,\ 0.5)$$

Solution 8: b. Testing whether the median change in the test score is positive (sign test)

Step 1: Define the hypotheses.

We test whether the median change in the test score is positive:

$$H_0 : \text{Median}_X \leq 0$$

$$H_A : \text{Median}_X > 0$$

$$\alpha = 0.05$$

Step 2: State the estimator.

For the sign test, we use the estimator B:

$$B = \text{number of observations where } X_i - 0 > 0$$

Under H_0, B is exactly Binom:

$$B \sim \text{Binom}(n,\ 0.5)$$

Step 3: Find the test value.

None of the changes equal 0, so all $n = 20$ observations are kept. In the frequency table, the Cumulative Percent column shows that 30.0% of the students, i.e. 6 of the 20, have a negative

change. So the other 14 have a positive change:

$$b = 20 - 6 = 14$$

Step 4: Find the p-value.

Since H_A says the median is positive, we need the probability of a value at least as large as b:

$$p = P(B \geq 14) = 1 - \texttt{binomcdf}(20,\ 0.5,\ 13) \approx 0.0577$$

Step 5: Make the decision.

The p-value is large, $p > 0.05$, so we cannot reject H_0.

Here we also have to check the assumption of the sign test:

- The variable is quantitative or ordinal.

The change in test score is quantitative, so this assumption is satisfied.

Answer: We cannot conclude, with 95% confidence, that the median change in the test score is positive - the median change could be negative or 0.

Tip: Wilcoxon Signed-Rank Test

The **Wilcoxon signed-rank test** is a non-parametric test used when data is ordinal or non-normal quantitative. It is used in two cases:

1. Compare a population median of one sample to a fixed value med_0.

$$H_0 : \text{median} = med_0 \qquad H_A : \text{median} \neq med_0$$

2. Compare the population medians of two paired samples.

$$H_0 : \text{median}_x = \text{median}_y \qquad H_A : \text{median}_x \neq \text{median}_y$$

It comes with two assumptions:

- The variable is quantitative or ordinal.
- $n > 20$.

We prefer it over the sign test, because it uses the size of each deviation, not just its sign.

Tip: Signed-Rank Test: Mechanics

In order to do the Wilcoxon signed-rank test, we use the estimator B. Here are the steps to find it, and its distribution under H_0:

1. Compute $d_i = |x_i - med_0|$ for every observation.
2. Sort the observations by d_i from smallest to largest, and assign each one a rank equal to its position in that order (the smallest gets rank 1, the next smallest rank 2, and so on).
   2.1 Ties get the average rank (e.g. if the 5th and 6th are tied, both get 5.5).
   2.2 If $d_i = 0$, drop that observation.
3. Calculate the estimator:

$$B = \text{sum of ranks for observations where } x_i - med_0 > 0$$

Under H_0:

$$B \approx \mathcal{N}\left(E(B) = \frac{n(n+1)}{4},\ Var(B) = \frac{n(n+1)(2n+1)}{24}\right)$$

Solution 8: c. Testing whether the median change in the test score is positive (signed-rank test)

Step 1: Define the hypotheses.

We test whether the median change in the test score is positive:

$$H_0 : \text{Median}_X \leq 0$$

$$H_A : \text{Median}_X > 0$$

$$\alpha = 0.05$$

Step 2: State the estimator.

For the Wilcoxon signed-rank test, we use the estimator B:

$$B = \text{sum of ranks for observations where } X_i - 0 > 0$$

Under H_0, B is approximately Normal:

$$B \approx \mathcal{N}\left(E(B) = \frac{n(n+1)}{4},\ Var(B) = \frac{n(n+1)(2n+1)}{24}\right)$$

Step 3: Find the test value.

For each observation, we take the absolute value of the change and rank these from smallest to largest. None of the changes equal zero, so all 20 observations are kept.

Many observations share the same absolute value, so each group of tied observations is given the average of the ranks they would otherwise occupy. For example, four observations have $|X| = 1$ (two of -1 and two of 1), so they occupy ranks 1 to 4 and each gets rank 2.5. Repeating this for every value of $|X|$ gives:

$\lvert X\rvert$	observations	ranks occupied	rank	positive changes
1	4	1 to 4	2.5	2
2	4	5 to 8	6.5	3
3	5	9 to 13	11	4
4	3	14 to 16	15	2
5	3	17 to 19	18	2
7	1	20	20	1

The observed sum of positive ranks, counting each rank once for every positive change that gets

it:

$$b = 2(2.5) + 3(6.5) + 4(11) + 2(15) + 2(18) + 1(20) = 154.5$$

With $n = 20$:

$$E(B) = \frac{(20)(21)}{4} = 105.0000$$

$$Var(B) = \frac{(20)(21)(41)}{24} = 717.5000$$

$$SD(B) = \sqrt{717.5000} = 26.7862$$

Step 4: Find the p-value.

Since H_A says the median is positive, we need the probability of a value at least as large as b:

$$p = P(B \geq 154.5) = \texttt{normalcdf}(\texttt{154.5, 1E99, 105, 26.7862}) \approx 0.0323$$

Step 5: Make the decision.

The p-value is small, $p < 0.05$, so we reject H_0.

Here we also have to check the assumptions of the signed-rank test:

- The variable is quantitative or ordinal.
- $n > 20$.

The change in test score is quantitative, so the first assumption is satisfied. Here $n = 20$, so the second assumption is right at the boundary of being satisfied.

Answer: We conclude with 95% confidence that the median change in the test score is positive. With $n = 20$, the sample-size assumption is right at the boundary of being satisfied.

Question 9

We want to test whether 14 boys and 13 girls differ in the time they need to solve the exam of statistical modelling. The data below show the order in which boys (M) and girls (F) handed in the exam.

gender	rank
F	1
F	2
M	3
F	4
F	5
M	6
M	7
F	8
F	9
F	10
M	11
F	12
M	13
F	14
F	15
F	16
M	17
M	18
F	19
M	20
F	21
M	22
M	23
M	24
M	25
M	26
M	27

a. Suppose analysis of the data indicates that the distribution of the solution time has the same shape for boys and girls. Use a MWW test to test whether the median time to solve the exam is larger for boys than for girls.

b. Discuss whether the assumptions of the test are satisfied.

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Tip: Mann-Whitney-Wilcoxon Test

The **Mann-Whitney-Wilcoxon test** is a non-parametric test used to compare the population medians of two independent samples, when the data is ordinal or non-normal quantitative.

$$H_0 : \text{median}_1 = \text{median}_2 \qquad H_A : \text{median}_1 \neq \text{median}_2$$

It comes with four assumptions:

- The variable is quantitative or ordinal.
- The two samples are independent.
- The two groups' distributions have the same shape (they can only differ by a shift in location).
- $n_1 + n_2 \geq 20$.

We prefer it over the independent-samples t-test when Normality/CLT fails, or the median is more meaningful than the mean.

Tip: Mann-Whitney-Wilcoxon Test: Mechanics

In order to do the Mann-Whitney-Wilcoxon test, we use the estimator W. Here are the steps to find it, and its distribution under H_0:

1. Pool both groups together and rank all observations from smallest to largest, ignoring group membership. Ties get the average rank.
2. Sum the ranks separately for each group.
3. Calculate the estimator:

   $$W = \text{the smaller of the two rank sums}$$

   with n_1 the size of the group that produced it, and n_2 the size of the other group.

Under H_0:

$$W \approx \mathcal{N}\left(E(W) = \frac{n_1(n_1+n_2+1)}{2},\ Var(W) = \frac{n_1 n_2(n_1+n_2+1)}{12}\right)$$

Solution 9: a. Testing whether the median time to solve the exam is larger for boys

Step 1: Define the hypotheses.

We test whether the median time to solve the exam is larger for boys than for girls:

$$H_0 : \text{Median}_{boys} \leq \text{Median}_{girls}$$

$$H_A : \text{Median}_{boys} > \text{Median}_{girls}$$

$$\alpha = 0.05$$

Step 2: State the estimator.

Boys and girls are two independent samples of different students, not the same students measured twice, and we only know the order in which they handed in the exam, i.e. the ranks of their solution times. We are asked about the median, so we use the non-parametric Mann-Whitney-Wilcoxon test to compare the two groups' medians.

For the Mann-Whitney-Wilcoxon test, we use the estimator W, the smaller of the two groups' rank sums:

$$W = \min(\text{rank sum}_{girls},\ \text{rank sum}_{boys})$$

Under H_0, W is approximately Normal:

$$W \approx \mathcal{N}\left(E(W) = \frac{n_1(n_1+n_2+1)}{2},\ Var(W) = \frac{n_1 n_2(n_1+n_2+1)}{12}\right)$$

Step 3: Find the test value.

The ranks are already given in the table above: the student who handed in first, i.e. with the

shortest solution time, has rank 1. Grouping them by gender:

$$\text{girls} : 1, 2, 4, 5, 8, 9, 10, 12, 14, 15, 16, 19, 21$$

$$\text{boys} : 3, 6, 7, 11, 13, 17, 18, 20, 22, 23, 24, 25, 26, 27$$

Summing the ranks separately for each group:

$$\begin{aligned}\text{rank sum}_{girls} &= 1 + 2 + 4 + 5 + 8 + 9 + 10 \\ &\quad + 12 + 14 + 15 + 16 + 19 + 21 = 136\end{aligned}$$

$$\begin{aligned}\text{rank sum}_{boys} &= 3 + 6 + 7 + 11 + 13 + 17 + 18 \\ &\quad + 20 + 22 + 23 + 24 + 25 + 26 + 27 = 242\end{aligned}$$

The girls have the smaller rank sum, so $w = 136$, with $n_1 = 13$ (girls) and $n_2 = 14$ (boys):

$$E(W) = \frac{(13)(28)}{2} = 182.0000$$

$$Var(W) = \frac{(13)(14)(28)}{12} = 424.6667$$

$$SD(W) = \sqrt{424.6667} = 20.6074$$

Step 4: Find the p-value.

Since H_A says the median is larger for boys, i.e. smaller for girls, we need the probability of a value at least as small as w:

$$p = P(W \leq 136) = \texttt{normalcdf}(-\texttt{1E99},\ \texttt{136},\ \texttt{182},\ \texttt{20.6074}) \approx 0.0128$$

Step 5: Make the decision.

The p-value is small, $p < 0.05$, so we reject H_0.

Answer: We conclude with 95% confidence that the median time to solve the exam is larger

for boys than for girls.

Solution 9: b. Discuss whether the assumptions of the test are satisfied

The Mann-Whitney-Wilcoxon test makes four assumptions:

- The variable is quantitative or ordinal.
- The two samples are independent.
- The two groups' distributions have the same shape.
- $n_1 + n_2 \geq 20$.

Step 1: Check the variable type.
The time to solve the exam is quantitative, and we use its ranks, so this assumption is satisfied.

Step 2: Check independence.
Boys and girls are different students, not the same students measured twice, so the two samples are independent. This assumption is satisfied.

Step 3: Check the same-shape assumption.
The question tells us that the distribution of the solution time has the same shape for boys and girls, so this assumption is satisfied.

Step 4: Check the sample-size assumption.
Here $n_1 + n_2 = 13 + 14 = 27 \geq 20$, so this assumption is satisfied.

Answer: All four assumptions of the Mann-Whitney-Wilcoxon test are satisfied.

Question 10

On 1/1/2017 the proportion of inhabitants of Flanders with a specific civil status is as follows:

Civil status	Population Proportion
Not married	.476
Married	.374
widowed	.060
Divorced	.090

In a random sample of 400 inhabitants of the population we obtain the following counts:

Civil status	Observed count in sample
Not married	175
Married	150
widowed	33
Divorced	42

a. Use an appropriate statistical test to investigate whether the sample is representative with respect to civil status.

b. Discuss whether the assumptions of the test are satisfied.

Tip: Chi-Square Goodness-of-Fit Test

The **chi-square goodness-of-fit test** is a non-parametric test used to test whether one sample, with a categorical variable of k categories, comes from a population with a fully specified, hypothesized distribution.

$$H_0 : \pi_1, \ldots, \pi_k \text{ are the hypothesized population proportions} \qquad H_A : H_0 \text{ is false}$$

It comes with two assumptions:

- All expected counts are larger than 1.
- At most 20% of the categories have an expected count smaller than 5.

Although it uses the same χ^2 statistic as the chi-square test for proportions/independence, this

is a different test: it compares one sample against a hypothesized distribution given directly by the exercise, not two or more groups against each other.

Tip: Chi-Square Goodness-of-Fit Test: Mechanics

In order to do the chi-square goodness-of-fit test, we use the estimator χ^2. Here are the steps to find it, and its distribution under H_0:

1. Build the table of observed counts O for every category.
2. Compute the expected count for every category, assuming H_0 is true:

$$E_k = n \times \pi_k$$

3. Calculate the estimator:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Under H_0:

$$\chi^2 \sim \chi^2_{k-1}$$

with k the number of categories.

Solution 10: a. Testing whether the sample is representative with respect to civil status

Step 1: Define the hypotheses.

Let π_1, π_2, π_3 and π_4 be the population proportions of each civil status:

$$\pi_1 = \text{proportion of inhabitants who are not married}$$

$$\pi_2 = \text{proportion of inhabitants who are married}$$

$$\pi_3 = \text{proportion of inhabitants who are widowed}$$

$$\pi_4 = \text{proportion of inhabitants who are divorced}$$

We test whether the sample is representative, i.e. whether it comes from a population with exactly the proportions given in the first table:

$$H_0 : \pi_1 = 0.476 \text{ and } \pi_2 = 0.374 \text{ and } \pi_3 = 0.060 \text{ and } \pi_4 = 0.090$$

$$H_A : H_0 \text{ is wrong}$$

$$\alpha = 0.05$$

Step 2: State the estimator.

There is only one sample here (the 400 inhabitants), split into $k = 4$ categories (the civil status of each inhabitant), and we are comparing it to a fully specified, hypothesized distribution across those categories, not to a second sample. This is exactly what the chi-square goodness-of-fit test is for.

For the chi-square goodness-of-fit test, we use the estimator χ^2:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Under H_0, χ^2 has a chi-square distribution with $k - 1$ degrees of freedom.

Step 3: Find the test value.

None of the expected counts are given, so we compute all four of them with $n = 400$:

$$E_1 = 400 \times 0.476 = 190.4 \quad E_2 = 400 \times 0.374 = 149.6$$

$$E_3 = 400 \times 0.060 = 24.0 \quad E_4 = 400 \times 0.090 = 36.0$$

With every expected count now known, we compute χ^2:

$$\chi^2 = \frac{(175 - 190.4)^2}{190.4} + \frac{(150 - 149.6)^2}{149.6} + \frac{(33 - 24)^2}{24} + \frac{(42 - 36)^2}{36} \approx 5.6217$$

with $df = k - 1 = 4 - 1 = 3$.

Step 4: Find the p-value.

The chi-square test is always right-tailed, whatever H_A says, since only a large χ^2 value counts as evidence against H_0:

$$p = P(\chi^2 \geq 5.6217) = \chi^2\texttt{cdf}(\texttt{5.6217, 1E99, 3}) \approx 0.1315$$

Step 5: Make the decision.

The p-value is large, $p > 0.05$, so we cannot reject H_0.

Answer: We cannot conclude, with 95% confidence, that the sample is not representative with respect to civil status - the sample could be representative.

Solution 10: b. Discuss whether the assumptions of the test are satisfied

The chi-square goodness-of-fit test makes two assumptions:

- All expected counts are larger than 1.
- At most 20% of the categories have an expected count smaller than 5.

Step 1: Check that all expected counts are larger than 1.

The smallest expected count is 24.0 (widowed), which is much larger than 1. This assumption is satisfied.

Step 2: Check that at most 20% of the categories have an expected count smaller than 5.

None of the four expected counts are smaller than 5 (0% of the categories), so this assumption is satisfied too.

Answer: Both assumptions of the chi-square goodness-of-fit test are satisfied.

Question 11

A sample of $n = 12$ male NBA basketball players have the following lengths (in cm):

$$183, 188, 192, 193, 200, 204, 206, 206, 206, 187, 194, 202$$

a. Conduct an appropriate statistical test to investigate whether the median length of NBA basketball players is larger than 190 cm. Make sure the assumptions of the test are satisfied.

b. Which test would you use if the sample size was $n = 30$? Justify your answer.

Tip: Sign Test

The **sign test** is a non-parametric test used when data is ordinal or quantitative. It is used in two cases:

1. Compare a population median of one sample to a fixed value med_0.

$$H_0 : \text{median} = med_0 \qquad H_A : \text{median} \neq med_0$$

2. Compare the population medians of two paired samples.

$$H_0 : \text{median}_x = \text{median}_y \qquad H_A : \text{median}_x \neq \text{median}_y$$

It comes with one assumption:

- The variable is quantitative or ordinal.

It is less powerful than the signed-rank test, because it only uses the direction of each deviation, not its size. We use it instead of the signed-rank test whenever $n \leq 20$, since the signed-rank test's Normal approximation is not valid below that threshold, while the sign test's Binom distribution is exact for any n.

Tip: Sign Test: Mechanics

In order to do the sign test, we use the estimator B. Here are the steps to find it, and its distribution under H_0:

1. Compute the deviation $x_i - med_0$ for every observation.
2. Drop every observation where that deviation is exactly 0. Let n be the number of observations left.
3. Calculate the estimator:

$$B = \text{number of observations where } x_i - med_0 > 0$$

Under H_0:

$$B \sim \text{Binom}(n,\ 0.5)$$

Solution 11: a. Testing whether the median length of NBA players is larger than 190 cm

Step 1: Define the hypotheses.

We test whether the median length of NBA basketball players is larger than 190 cm:

$$H_0 : \text{Median}_{length} \leq 190$$

$$H_A : \text{Median}_{length} > 190$$

$$\alpha = 0.05$$

Step 2: State the estimator.

The 12 players form a single sample, and we are comparing their median length to a fixed value ($med_0 = 190$), so this is a one-sample test against a fixed value. The question is about the median, so we need a non-parametric test. The signed-rank test requires $n > 20$, but here $n = 12$, so we use the sign test, which is valid for any n.

For the sign test, we use the estimator B:

$$B = \text{number of observations where } length_i - 190 > 0$$

Under H_0, B is exactly Binom:

$$B \sim \text{Binom}(n,\ 0.5)$$

Step 3: Find the test value.

None of the lengths equal 190, so all $n = 12$ observations are kept. Counting the observations where $length_i > 190$ (namely $192, 193, 200, 204, 206, 206, 206, 194, 202$):

$$b = 9$$

Step 4: Find the p-value.

Since H_A says the median is larger, we need the probability of a value at least as large as b:

$$p = P(B \geq 9) = 1 - \texttt{binomcdf}(12,\ 0.5,\ 8) \approx 0.0730$$

Step 5: Make the decision.

The p-value is large, $p > 0.05$, so we cannot reject H_0.

Here we also have to check the assumption of the sign test:

- The variable is quantitative or ordinal.

The length of a player is quantitative, so this assumption is satisfied.

Answer: We cannot conclude, with 95% confidence, that the median length of NBA basketball players is larger than 190 cm - the median could be 190 cm or smaller.

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Tip: Wilcoxon Signed-Rank Test

The **Wilcoxon signed-rank test** is a non-parametric test used when data is ordinal or non-normal quantitative. It is used in two cases:

1. Compare a population median of one sample to a fixed value med_0.

$$H_0 : \text{median} = med_0 \qquad H_A : \text{median} \neq med_0$$

2. Compare the population medians of two paired samples.

$$H_0 : \text{median}_x = \text{median}_y \qquad H_A : \text{median}_x \neq \text{median}_y$$

It comes with two assumptions:

- The variable is quantitative or ordinal.
- $n > 20$.

We prefer it over the sign test, because it uses the size of each deviation, not just its sign.

Solution 11: b. Which test to use if the sample size was $n = 30$

The test is still a one-sample test of the median against a fixed value ($med_0 = 190$), so we still need a non-parametric test for one sample.

With $n = 30 > 20$, the signed-rank test can be used, and it is the best choice because it uses the size of each player's deviation from 190, not just its direction. The sign test would still be valid, but it is less powerful than the signed-rank test.

Answer: The Wilcoxon signed-rank test, since $n = 30 > 20$ and it is more powerful than the sign test.

Question 12

In a survey on transportation a random sample of 661 respondents indicate to what extent they have experience with biking (1=no experience, 2=little experience, 3=very experienced). The following cross-table shows the relation between **gender** and the categorical variable **biking_experience**.

gender * biking_experience Crosstabulation

		biking_experience			
		no experience	little experience	very experienced	Total
male	Count	51	30	193	274
	Expected Count	66.7	43.1	A	274.0
female	Count	110	74	203	387
	Expected Count	94.3	60.9	231.8	387.0
Total	Count	161	104	396	661
	Expected Count	161.0	104.0	396.0	661.0

a. Compute the value A in the cross-table.

b. Use an appropriate statistical test to investigate whether the distribution of the categorical variable **biking_experience** is different for males and females.

c. Discuss whether the assumptions of the test are satisfied.

d. Suppose in the sample, the age of the respondents (measured in years) ranges from 16 to 81. Which test can you conduct to test the null hypothesis that, in the population, the average age is the same at the three biking experience levels? Discuss the assumptions that need to be satisfied for this test to be valid.

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Tip: Chi-Square Test for Proportions

The **chi-square test** is a non-parametric test used when the outcome is categorical (nominal), not quantitative. It is used to test whether the proportions of a categorical outcome differ between two or more independent samples.

$$H_0 : \pi_1 = \pi_2 = \dots \quad \text{(population proportions are equal)}$$

$$H_A : \text{at least one } \pi_i \text{ is different}$$

It comes with two assumptions:

- All expected counts are larger than 1.
- At most 20% of the cells have an expected count smaller than 5.

Tip: Chi-Square Test: Mechanics

In order to do the chi-square test, we use the estimator χ^2. Here are the steps to find it, and its distribution under H_0:

1. Build the table of observed counts O, with the groups as rows and the outcome categories as columns (or vice versa).
2. Compute the expected count for every cell, assuming H_0 is true:
$$E = \frac{\text{row total} \times \text{column total}}{\text{grand total}}$$
3. Calculate the estimator:
$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Under H_0:

$$\chi^2 \sim \chi^2_{(r-1)(k-1)}$$

with r the number of rows and k the number of columns in the table.

Solution 12: a. Compute the value A in the cross-table

A is the expected count of males who are very experienced with biking, assuming H_0 is true. It is the row total of males times the column total of very experienced respondents, divided by the grand total:

$$A = E_{3|M} = \frac{274 \times 396}{661} \approx 164.2$$

Answer: $A \approx 164.2$

Solution 12: b. Testing whether the distribution of biking experience differs between males and females

Step 1: Define the hypotheses.

Let $\pi_{i|M}$ and $\pi_{i|F}$ be the proportion of males, respectively females, with biking experience level i:

$$\pi_{1|M} = \text{proportion of males who have no experience}$$

$$\pi_{2|M} = \text{proportion of males who have little experience}$$

$$\pi_{3|M} = \text{proportion of males who are very experienced}$$

$$\pi_{1|F} = \text{proportion of females who have no experience}$$

$$\pi_{2|F} = \text{proportion of females who have little experience}$$

$$\pi_{3|F} = \text{proportion of females who are very experienced}$$

We test whether the population distribution of biking experience is the same for males and

females:

$$H_0 : \pi_{1|M} = \pi_{1|F} \text{ and } \pi_{2|M} = \pi_{2|F} \text{ and } \pi_{3|M} = \pi_{3|F}$$

$$H_A : H_0 \text{ is wrong}$$

$$\alpha = 0.05$$

Biking experience has three categories, not two, so one equality alone would not fully describe "the same distribution" - we need every category's proportion to match between males and females.

Step 2: State the estimator.

Male and female respondents are two independent groups of people, not the same people measured twice, and biking experience is a categorical variable (no/little/very experienced), not quantitative. This is exactly what the chi-square test for proportions is for: comparing a categorical outcome across two or more independent groups.

For the chi-square test, we use the estimator χ^2:

$$\chi^2 = \sum \frac{(O-E)^2}{E}, \qquad E = \frac{\text{row total} \times \text{column total}}{\text{grand total}}$$

Under H_0, χ^2 has a chi-square distribution with $(r-1)(k-1)$ degrees of freedom.

Step 3: Find the test value.

The expected counts are already shown in the crosstable above (with A from part a); computing them to more decimal places:

$$E_{1|M} = \frac{274 \times 161}{661} \approx 66.7383 \quad E_{1|F} = \frac{387 \times 161}{661} \approx 94.2617$$

$$E_{2|M} = \frac{274 \times 104}{661} \approx 43.1104 \quad E_{2|F} = \frac{387 \times 104}{661} \approx 60.8896$$

$$E_{3|M} = \frac{274 \times 396}{661} \approx 164.1513 \quad E_{3|F} = \frac{387 \times 396}{661} \approx 231.8487$$

With every expected count now known, we compute χ^2:

$$\chi^2 = \frac{(51 - 66.7383)^2}{66.7383} + \cdots + \frac{(203 - 231.8487)^2}{231.8487} \approx 21.8087$$

with $df = (2-1)(3-1) = 2$.

Step 4: Find the p-value.

The chi-square test is always right-tailed, whatever H_A says, since only a large χ^2 value counts as evidence against H_0:

$$p = P(\chi^2 \geq 21.8087) = \chi^2\texttt{cdf}(\texttt{21.8087, 1E99, 2}) \approx 0.0000$$

Step 5: Make the decision.

The p-value is small, $p < 0.05$, so we reject H_0.

Answer: We conclude with 95% confidence that the distribution of biking experience is different for males and females.

Solution 12: c. Discuss whether the assumptions of the test are satisfied

The chi-square test makes two assumptions:

- All expected counts are larger than 1.
- At most 20% of the cells have an expected count smaller than 5.

Step 1: Check that all expected counts are larger than 1.

The six expected counts are 66.7383, 43.1104, 164.1513, 94.2617, 60.8896, and 231.8487, all larger than 1. This assumption is satisfied.

Step 2: Check that at most 20% of the cells have an expected count smaller than 5.

None of the six expected counts are smaller than 5 (0% of the cells), so this assumption is

satisfied too.

Answer: Both assumptions of the chi-square test are satisfied.

Tip: ANOVA

ANOVA is used to test whether the population mean of a quantitative variable is equal across more than 2 independent samples, defined by a categorical variable.

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_G \qquad H_A : \text{at least one } \mu_g \text{ is different}$$

It comes with three assumptions:

1. Observations are independent.
2. Model errors are Normally distributed. With a large sample this is not needed, since the CLT ensures the test statistic has an approximate F-distribution.
3. Error variances are equal across groups.

Solution 12: d. Test and assumptions for comparing average age across biking experience levels

We have more than 2 independent groups here, the 3 biking experience levels, and we want to compare the population mean of age, a quantitative variable, across them. That means we use ANOVA.

The test comes with three assumptions:

1. Observations are independent.

2. Model errors are Normally distributed.
3. Error variances are equal across groups.

Step 1: Check independence.
Since the 661 respondents form a random sample, we can simply assume that this assumption holds.

Step 2: Check Normality of the model errors.
We can check this with a Normality test, either Kolmogorov-Smirnov or Shapiro-Wilk.

Here the sample is large ($n = 661$), so even if this assumption were violated it would not really be a problem: the central limit theorem still guarantees that ANOVA's F test statistic has an approximately correct F-distribution.

Step 3: Check equal variances.
We can use the test of Levene, which tests whether the error variance of age is equal across the 3 biking experience levels.

Answer: We can use one-way ANOVA. Its assumptions are independent observations (satisfied by the random sample), Normal model errors (checked with a Kolmogorov-Smirnov or Shapiro-Wilk test, and not a problem here since $n = 661$ is large), and equal error variances (checked with the test of Levene).

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Question 13

In a survey about ecotourism 110 respondents answered the following question:

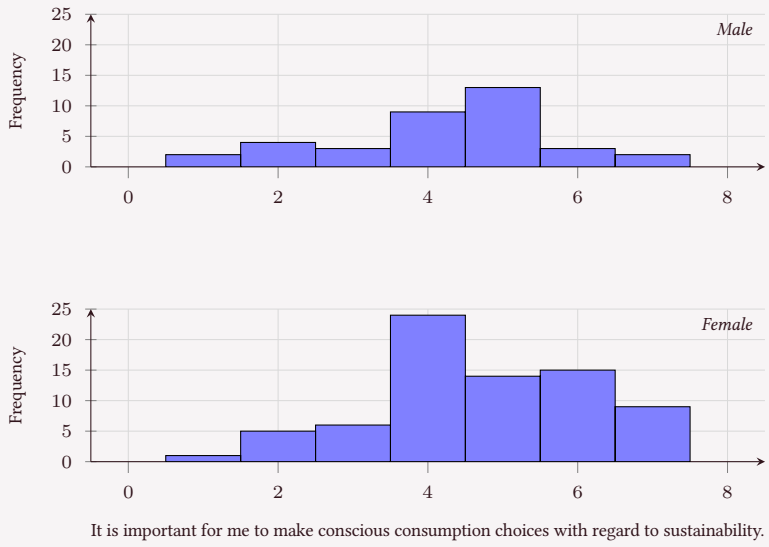
> It is important for me to make conscious consumption choices with respect to sustainability (1=strongly disagree, ..., 7=strongly agree).

We use a Mann-Whitney-Wilcoxon test to compare the median scores of males and females on the question. Analysis with SPSS yields the following results:

Report

It is important for me to make conscious consumption choices with regard to sustainability.

What is your gender?	Mean	N	Std. Deviation	Median
Male	4.22	36	1.495	4.50
Female	4.70	74	1.450	5.00
Total	4.55	110	1.475	5.00



Ranks

		N	Mean Rank	Sum of Ranks
It is important for me to make conscious consumption choices with regard to sustainability.	Male	36	49.85	1794.50
	Female	74	58.25	4310.50
	Total	110		

a. Test whether the median score on the question is higher for females than for males.

b. Discuss whether the assumptions of the test are satisfied.

Tip: Mann-Whitney-Wilcoxon Test

The **Mann-Whitney-Wilcoxon test** is a non-parametric test used to compare the population medians of two independent samples, when the data is ordinal or non-normal quantitative.

$$H_0 : \text{median}_1 = \text{median}_2 \qquad H_A : \text{median}_1 \neq \text{median}_2$$

It comes with four assumptions:

- The variable is quantitative or ordinal.
- The two samples are independent.
- The two groups' distributions have the same shape (they can only differ by a shift in location).
- $n_1 + n_2 \geq 20$.

We prefer it over the independent-samples t-test when Normality/CLT fails, or the median is more meaningful than the mean.

Tip: Mann-Whitney-Wilcoxon Test: Mechanics

In order to do the Mann-Whitney-Wilcoxon test, we use the estimator W. Here are the steps to find it, and its distribution under H_0:

1. Pool both groups together and rank all observations from smallest to largest, ignoring group membership. Ties get the average rank.
2. Sum the ranks separately for each group.

3. Calculate the estimator:

$$W = \text{the smaller of the two rank sums}$$

with n_1 the size of the group that produced it, and n_2 the size of the other group.

Under H_0:

$$W \approx \mathcal{N}\left(E(W) = \frac{n_1(n_1+n_2+1)}{2},\ Var(W) = \frac{n_1 n_2(n_1+n_2+1)}{12}\right)$$

Solution 13: a. Testing whether the median score is higher for females

Step 1: Define the hypotheses.

We test whether the median score on the question is higher for females than for males:

$$H_0 : \text{Median}_{females} \leq \text{Median}_{males}$$

$$H_A : \text{Median}_{females} > \text{Median}_{males}$$

$$\alpha = 0.05$$

Step 2: State the estimator.

Males and females are two independent samples of different respondents, not the same respondents measured twice, and the score on a 7-point scale is an ordinal variable. We are asked about the median, so we use the non-parametric Mann-Whitney-Wilcoxon test to compare the two groups' medians, instead of the independent-samples t-test.

For the Mann-Whitney-Wilcoxon test, we use the estimator W, the smaller of the two groups' rank sums:

$$W = \min(\text{rank sum}_{males},\ \text{rank sum}_{females})$$

Under H_0, W is approximately Normal:

$$W \approx \mathcal{N}\left(E(W) = \frac{n_1(n_1+n_2+1)}{2},\ Var(W) = \frac{n_1 n_2(n_1+n_2+1)}{12}\right)$$

Step 3: Find the test value.

SPSS already ranked all 110 scores together. In the Sum of Ranks column of the Ranks table, we read:

$$\text{rank sum}_{males} = 1794.5$$

$$\text{rank sum}_{females} = 4310.5$$

The males have the smaller rank sum, so $w = 1794.5$, with $n_1 = 36$ (males) and $n_2 = 74$ (females):

$$E(W) = \frac{(36)(111)}{2} = 1998.0000$$

$$Var(W) = \frac{(36)(74)(111)}{12} = 24642.0000$$

$$SD(W) = \sqrt{24642.0000} = 156.9777$$

Step 4: Find the p-value.

Since H_A says the median is higher for females, i.e. lower for males, we need the probability of a value at least as small as w:

$$p = P(W \leq 1794.5) = \texttt{normalcdf}(-\texttt{1E99},\ \texttt{1794.5},\ \texttt{1998},\ \texttt{156.9777}) \approx 0.0974$$

Step 5: Make the decision.

The p-value is large, $p > 0.05$, so we cannot reject H_0.

Answer: We cannot conclude, with 95% confidence, that the median score on the question is higher for females than for males - the females' median could be smaller than or equal to the males' median.

Solution 13: b. Discuss whether the assumptions of the test are satisfied

The Mann-Whitney-Wilcoxon test makes four assumptions:

- The variable is quantitative or ordinal.
- The two samples are independent.
- The two groups' distributions have the same shape.
- $n_1 + n_2 \geq 20$.

Step 1: Check the variable type.
The score on a scale from 1 to 7 is ordinal, so this assumption is satisfied.

Step 2: Check independence.
Males and females are different respondents, not the same respondents measured twice, so the two samples are independent. This assumption is satisfied.

Step 3: Check the same-shape assumption.
Looking at the two histograms, both distributions have one peak in the middle of the scale, at 5 for males and at 4 for females, with fewer scores at both ends. The standard deviations are also very similar (1.495 for males and 1.450 for females). The two distributions look reasonably similar in shape, so this assumption is roughly satisfied.

Step 4: Check the sample-size assumption.
Here $n_1 + n_2 = 36 + 74 = 110 \geq 20$, so this assumption is satisfied.

Answer: All four assumptions of the Mann-Whitney-Wilcoxon test are satisfied (the same-shape assumption roughly).

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Question 14

A factory wants to investigate whether the number of products produced by a machine in a period of an hour is Normally distributed. The variable **production** in the table below indicates the number of products produced for a random sample of 11 hours. The sample mean and sample standard deviation of the variable production equal $\bar{x} = 98.182$ and $s_x = 14.11253$. The table below also shows part of the computations needed for the Kolmogorov-Smirnov (KS) test statistic.

obs	production	EDF	CDF	EDF-CDF	EDF-CDF+
1	80	0.0909	0.0988	-0.0079	-0.1232
2	87	0.2727	0.2141	0.0586	0.0586
3	87	0.2727	0.2141	0.0586	-0.0327
4	91	0.3636	0.3054	0.0582	0.0330
5	92	0.4545	0.3307	0.1239	0.0711
6	94	0.5455	0.3835	0.1620	0.0506
7	98	0.7273	0.4949	0.2324	0.2324
8	98	A	B	C	D
9	108	E	F	G	H
10	119	0.9091	0.9299	-0.0208	-0.0666
11	126	1.0000	0.9756	0.0244	1.0000

a. Compute the numbers in the cells A, B, C, D, E, F, G and H.

b. Compute the value of the KS test statistic.

Tip: Kolmogorov-Smirnov Test

The Kolmogorov-Smirnov (K-S) test is a non-parametric goodness-of-fit test used to test whether one sample, with a quantitative variable, comes from a population that is Normally distributed.

H_0 : the population distribution is Normal $\qquad$ H_A : the population distribution is not Normal

The test compares the empirical distribution function (EDF) of the sample to the cumulative distribution function (CDF) of that Normal distribution: the larger the distance between them, the less plausible H_0 is.

Tip: Kolmogorov-Smirnov Test: Mechanics

In order to do the Kolmogorov-Smirnov test, we compare the EDF and the CDF at every observation. Here are the steps:

1. Sort the observations from smallest to largest.
2. Compute the empirical distribution function and the cumulative distribution function:

$$EDF_i = \hat{F}(x_i) = \frac{i}{n}$$

$$CDF_i = F(x_i) = P(X \leq x_i) = \texttt{normalcdf}(-\texttt{1E99}, \texttt{x}_\texttt{i}, \bar{\texttt{x}}, \texttt{s})$$

   with i the position of x_i in the sorted list. Tied values each keep their own position, so they get a different EDF but the same CDF.
3. Calculate the following two values for all observations:

$$EDF - CDF = EDF_i - CDF_i$$

$$EDF - CDF^+ = EDF_i - CDF_{i+1}$$

   The last observation has no next one, so $EDF - CDF^+$ is left blank for it.
4. Find the estimator D: the largest absolute value among the values from the previous step:

$$D = \max\left(|EDF - CDF|,\ |EDF - CDF^+|\right)$$

For example, we illustrate this on a small sample of 5 observations:

obs	x	EDF	CDF	EDF−CDF	EDF−CDF$^+$
1	42	0.2000	0.1160	0.0840	-0.1029
2	47	0.4000	0.3029	0.0971	-0.0034
3	49	0.6000	0.4034	0.1966	-0.1158
4	55	0.8000	0.7158	0.0842	-0.1170
5	61	1.0000	0.9170	0.0830	

At the third observation ($x = 49$) we find:

$$EDF - CDF = 0.6000 - 0.4034 = 0.1966$$

$$EDF - CDF^{+} = 0.6000 - 0.7158 = -0.1158$$

and, using every value in the table:

$$D = \max\left(|EDF - CDF|,\ |EDF - CDF^{+}|\right) = 0.1966$$

Solution 14: a. Compute A, B, C, D, E, F, G and H

The sample mean and standard deviation of production are given:

$$\overline{x} = 98.182$$

$$s = 14.11253$$

Step 1: Find A and E.

A and E are the EDF at the 8th and 9th observations:

$$A = EDF_8 = \frac{8}{11} = 0.7273 \qquad E = EDF_9 = \frac{9}{11} = 0.8182$$

Step 2: Find B and F.

B and F are the CDF at the 8th and 9th observations:

$$B = CDF_8 = P(X \leq 98) = \texttt{normalcdf}(-\texttt{1E99},\ \texttt{98},\ \texttt{98.182},\ \texttt{14.11253}) = 0.4949$$

$$F = CDF_9 = P(X \leq 108) = \texttt{normalcdf}(-\texttt{1E99},\ \texttt{108},\ \texttt{98.182},\ \texttt{14.11253}) = 0.7567$$

Step 3: Find C and D.

C is $EDF - CDF$ at the 8th observation, and D is $EDF - CDF^{+}$ at the 8th observation,

comparing it to the CDF at the *next* observation ($x = 108$):

$$C = EDF_8 - CDF_8 = 0.7273 - 0.4949 = 0.2324$$

$$D = EDF_8 - CDF_9 = 0.7273 - 0.7567 = -0.0294$$

Step 4: Find G and H.

G is $EDF - CDF$ at the 9th observation, and H is $EDF - CDF^+$ at the 9th observation, comparing it to the CDF at the *next* observation ($x = 119$):

$$G = EDF_9 - CDF_9 = 0.8182 - 0.7567 = 0.0615$$

$$H = EDF_9 - CDF_{10} = 0.8182 - 0.9299 = -0.1117$$

Note that the given table gives tied values the same EDF (both 87s get 0.2727). That changes nothing here: A belongs to the second 98, so $A = 8/11$ either way.

Answer:

A = 0.7273 E = 0.8182
B = 0.4949 F = 0.7567
C = 0.2324 G = 0.0615
D = −0.0294 H = −0.1117

Solution 14: b. Compute the value of the KS test statistic

It is the largest absolute value among every $EDF - CDF$ and $EDF - CDF^+$ in the table. The last observation has no next one, so its $EDF - CDF^+$ value is not used:

$$\max\left(|EDF - CDF|,\ |EDF - CDF^+|\right) = 0.2324$$

which is the C value at the 8th observation (and the same value at the 7th observation).

Answer: Kolmogorov-Smirnov statistic $= 0.2324$

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Question 15

12 students obtain the following grade for the exam of statistical modelling in September.

$$6,\ 7,\ 8,\ 9,\ 9,\ 11,\ 11,\ 11,\ 14,\ 16,\ 17,\ 20$$

Compute the value of the Kolmogorov-Smirnov test statistic for testing whether the grades come from a Normal distribution. Use for instance Excel to perform the calculations.

Tip: Kolmogorov-Smirnov Test

The Kolmogorov-Smirnov (K-S) test is a non-parametric goodness-of-fit test used to test whether one sample, with a quantitative variable, comes from a population that is Normally distributed.

H_0 : the population distribution is Normal $\qquad$ H_A : the population distribution is not Normal

The test compares the empirical distribution function (EDF) of the sample to the cumulative distribution function (CDF) of that Normal distribution: the larger the distance between them, the less plausible H_0 is.

Tip: Kolmogorov-Smirnov Test: Mechanics

In order to do the Kolmogorov-Smirnov test, we compare the EDF and the CDF at every observation. Here are the steps:

1. Sort the observations from smallest to largest.
2. Compute the empirical distribution function and the cumulative distribution function:

$$EDF_i = \hat{F}(x_i) = \frac{i}{n}$$

$$CDF_i = F(x_i) = P(X \leq x_i) = \texttt{normalcdf}(-\texttt{1E99},\ \texttt{x}_\texttt{i},\ \bar{\texttt{x}},\ \texttt{s})$$

with i the position of x_i in the sorted list. Tied values each keep their own position, so they get a different EDF but the same CDF.

3. Calculate the following two values for all observations:

$$EDF - CDF = EDF_i - CDF_i$$

$$EDF - CDF^+ = EDF_i - CDF_{i+1}$$

The last observation has no next one, so $EDF - CDF^+$ is left blank for it.

4. Find the estimator D: the largest absolute value among the values from the previous step:

$$D = \max\left(|EDF - CDF|,\ |EDF - CDF^+|\right)$$

For example, we illustrate this on a small sample of 5 observations:

obs	x	EDF	CDF	EDF−CDF	EDF−CDF$^+$
1	42	0.2000	0.1160	0.0840	-0.1029
2	47	0.4000	0.3029	0.0971	-0.0034
3	49	0.6000	0.4034	0.1966	-0.1158
4	55	0.8000	0.7158	0.0842	-0.1170
5	61	1.0000	0.9170	0.0830	

At the third observation ($x = 49$) we find:

$$EDF - CDF = 0.6000 - 0.4034 = 0.1966$$

$$EDF - CDF^+ = 0.6000 - 0.7158 = -0.1158$$

and, using every value in the table:

$$D = \max\left(|EDF - CDF|,\ |EDF - CDF^+|\right) = 0.1966$$

Solution 15: Compute the Kolmogorov-Smirnov statistic

With $n = 12$ observations, the sample mean and standard deviation of the grades are:

$$\overline{x} = \frac{\sum x_i}{n} = \frac{6 + 7 + \cdots + 20}{12} = \frac{139}{12} = 11.5833$$

$$s = \sqrt{\frac{1}{n-1}\sum (x_i - \overline{x})^2} = \sqrt{\frac{1}{11}\left((6 - 11.5833)^2 + \cdots + (20 - 11.5833)^2\right)} = 4.3161$$

Step 1: Sort the observations.

The grades are already sorted from smallest to largest, so the i-th grade in the list is observation i.

Step 2: Find the EDF and the CDF.

For every observation, $EDF_i = i/12$ and the CDF uses the sample mean and standard deviation. For example, at the 5th observation ($x = 9$):

$$EDF_5 = \frac{5}{12} = 0.4167$$

$$CDF_5 = P(X \leq 9) = \texttt{normalcdf}(-\texttt{1E99}, \texttt{ 9}, \texttt{ 11.5833}, \texttt{ 4.3161}) = 0.2747$$

Step 3: Find $EDF - CDF$ and $EDF - CDF^+$.

For example, at the 5th observation, comparing with the CDF at the *next* observation ($x = 11$) for $EDF - CDF^+$:

$$EDF - CDF = 0.4167 - 0.2747 = 0.1420$$

$$EDF - CDF^+ = 0.4167 - 0.4462 = -0.0295$$

Repeating this for every observation gives:

obs	grade	EDF	CDF	EDF−CDF	EDF−CDF$^+$
1	6	0.0833	0.0979	-0.0146	-0.0608
2	7	0.1667	0.1441	0.0226	-0.0365
3	8	0.2500	0.2032	0.0468	-0.0247
4	9	0.3333	0.2747	0.0586	0.0586
5	9	0.4167	0.2747	0.1420	-0.0295
6	11	0.5000	0.4462	0.0538	0.0538
7	11	0.5833	0.4462	0.1371	0.1371
8	11	0.6667	0.4462	0.2205	-0.0455
9	14	0.7500	0.7122	0.0378	-0.0969
10	16	0.8333	0.8469	-0.0136	-0.0620
11	17	0.9167	0.8953	0.0214	-0.0577
12	20	1.0000	0.9744	0.0256	

Step 4: Find the Kolmogorov-Smirnov statistic.

It is the largest absolute value among every $EDF - CDF$ and $EDF - CDF^+$ in the table:

$$D = \max\left(|EDF - CDF|,\ |EDF - CDF^+|\right) = 0.2205$$

which is the $EDF - CDF$ value at the 8th observation.

Answer: Kolmogorov-Smirnov statistic $= 0.2205$

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Question 16

A simple random sample of 250 lecturers who teach statistics indicate which type of software package they prefer, S1, S2, S3 or S4.

100 lecturers prefer S1, 80 lecturers prefer S2, 35 lecturers prefer S3 and 35 lecturers prefer S4.

Can you conclude from the data that lecturers prefer a certain software package more? Use $\alpha = 0.05$. Indicate whether the assumptions of the test are satisfied.

Tip: Chi-Square Goodness-of-Fit Test

The **chi-square goodness-of-fit test** is a non-parametric test used to test whether one sample, with a categorical variable of k categories, comes from a population with a fully specified, hypothesized distribution.

$$H_0 : \pi_1, \ldots, \pi_k \text{ are the hypothesized population proportions} \qquad H_A : H_0 \text{ is false}$$

It comes with two assumptions:

- All expected counts are larger than 1.
- At most 20% of the categories have an expected count smaller than 5.

Although it uses the same χ^2 statistic as the chi-square test for proportions/independence, this is a different test: it compares one sample against a hypothesized distribution given directly by the exercise, not two or more groups against each other.

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Tip: Chi-Square Goodness-of-Fit Test: Mechanics

In order to do the chi-square goodness-of-fit test, we use the estimator χ^2. Here are the steps to find it, and its distribution under H_0:

1. Build the table of observed counts O for every category.
2. Compute the expected count for every category, assuming H_0 is true:

$$E_k = n \times \pi_k$$

3. Calculate the estimator:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Under H_0:

$$\chi^2 \sim \chi^2_{k-1}$$

with k the number of categories.

Solution 16: Testing whether lecturers prefer a certain software package more

Step 1: Define the hypotheses.

Let π_1, π_2, π_3 and π_4 be the population proportions of lecturers who prefer S1, S2, S3 and S4. If lecturers do not prefer any package more, each package is preferred by a quarter of the lecturers:

$$H_0 : \pi_1 = 0.25 \text{ and } \pi_2 = 0.25 \text{ and } \pi_3 = 0.25 \text{ and } \pi_4 = 0.25$$

$$H_A : H_0 \text{ is wrong}$$

$$\alpha = 0.05$$

Step 2: State the estimator.

There is only one sample here (the 250 lecturers), split into $k = 4$ categories (the preferred software package of each lecturer), and we are comparing it to a fully specified, hypothesized

distribution across those categories, not to a second sample. This is exactly what the chi-square goodness-of-fit test is for.

For the chi-square goodness-of-fit test, we use the estimator χ^2:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Under H_0, χ^2 has a chi-square distribution with $k-1$ degrees of freedom.

Step 3: Find the test value.

None of the expected counts are given. With $n = 250$, every package has the same expected count:

$$E_1 = E_2 = E_3 = E_4 = 250 \times 0.25 = 62.5$$

With every expected count now known, we compute χ^2:

$$\chi^2 = \frac{(100-62.5)^2}{62.5} + \frac{(80-62.5)^2}{62.5} + \frac{(35-62.5)^2}{62.5} + \frac{(35-62.5)^2}{62.5} = 51.6000$$

with $df = k - 1 = 4 - 1 = 3$.

Step 4: Find the p-value.

The chi-square test is always right-tailed, whatever H_A says, since only a large χ^2 value counts as evidence against H_0:

$$p = P(\chi^2 \geq 51.6000) = \chi^2\texttt{cdf(51.6, 1E99, 3)} \approx 0.0000$$

Step 5: Make the decision.

The p-value is small, $p < 0.05$, so we reject H_0.

Here we also have to check the assumptions of the chi-square goodness-of-fit test:

- All expected counts are larger than 1.
- At most 20% of the categories have an expected count smaller than 5.

Every expected count equals 62.5, which is larger than 1, so the first assumption is satisfied.

None of the four expected counts are smaller than 5 (0% of the categories), so the second assumption is satisfied too.

Answer: We conclude with 95% confidence that lecturers do not prefer the four software packages equally - some packages are preferred more than others. Both assumptions of the test are satisfied.

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Question 17

A firm aims to evaluate whether the type of sales training affects the sales performance of managers. They randomly assign 10 sales managers to three conditions: 20 managers receive no sales training (=control condition), 10 managers receive sales training A and 10 managers receive sales training B. 6 weeks after the training the sales turnover generated by each manager is gathered.

a. Which kind of test can you conduct to test the global null hypothesis that the average turnover is equal across the three conditions?

b. Which kind of test can you use to test the difference in average sales performance for each pair of conditions?

Tip: ANOVA

ANOVA is used to test whether the population mean of a quantitative variable is equal across more than 2 independent samples, defined by a categorical variable.

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_G \qquad H_A : \text{at least one } \mu_g \text{ is different}$$

It comes with three assumptions:

1. Observations are independent.
2. Model errors are Normally distributed. With a large sample this is not needed, since the CLT ensures the test statistic has an approximate F-distribution.
3. Error variances are equal across groups.

Solution 17: a. Test for the global null hypothesis

We have more than 2 independent groups here, the three training conditions (no training, training A, training B), and we want to compare the population mean of sales turnover, a quantitative variable, across them. That means we use ANOVA, with sales turnover as the dependent variable and a categorical variable with the three conditions as levels.

Answer: We can use one-way ANOVA to test whether the average turnover differs across the three training conditions.

Tip: Tukey Post-Hoc Test

Once ANOVA finds that at least one group differs, a **Tukey post-hoc test** tells us specifically which pairs of groups differ. It is really just an independent-samples t-test, repeated for every pair of groups:

$$H_0 : \mu_i = \mu_j \qquad H_A : \mu_i \neq \mu_j$$

The problem is that running many tests back to back increases the chance that at least one of them rejects H_0 just by chance, even if every H_0 is actually true - the more tests we run, the more likely we are to make at least one mistake. Tukey corrects the p-values so that the probability of making at least one such mistake, across the whole set of comparisons, stays at $\alpha = 0.05$.

Solution 17: b. Test for comparing pairs of conditions

To test the difference in average sales performance for each pair of conditions, we can use a Tukey post-hoc test. This corrects the p-values of the pairwise comparisons for the inflation of the type-I error caused by running multiple tests.

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Answer: We can use a Tukey post-hoc test to test the difference in average sales performance for each pair of conditions.

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Question 18

A researcher is conducting an experiment to evaluate the impact of market context on strategic performance in a competitive setting. The researcher recruits 60 subjects from the entering class of MBA students at a large university. Each subject is put in charge of a firm in a market simulation game and is given the objective of strengthening the competitive position of his or her firm.

After 2 hours of play, the subject is given advice about the changing market context in the industry. Each subject is randomly assigned to one of three conditions in a balanced design. In the first control condition subjects receive no advice. In the second condition, the advice suggests to increase competition. In the third condition, the advice suggests to increase cooperation. Each subject then continues to play the game for 2 more hours.

The researcher uses the following measure of strategic performance:

Score_marketshare =

100 $\times$ market share after 4 hours of play / market share after 2 hours of play.

We conduct a one-way ANOVA with as dependent variable **score_marketshare** and as factor **condition**. Part of the SPSS output of the analysis is shown below:

ANOVA

score_marketshare

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	10290.700	2			
Within Groups	22982.900	57			
Total	33273.600	59			

Test of Homogeneity of Variances

score_marketshare

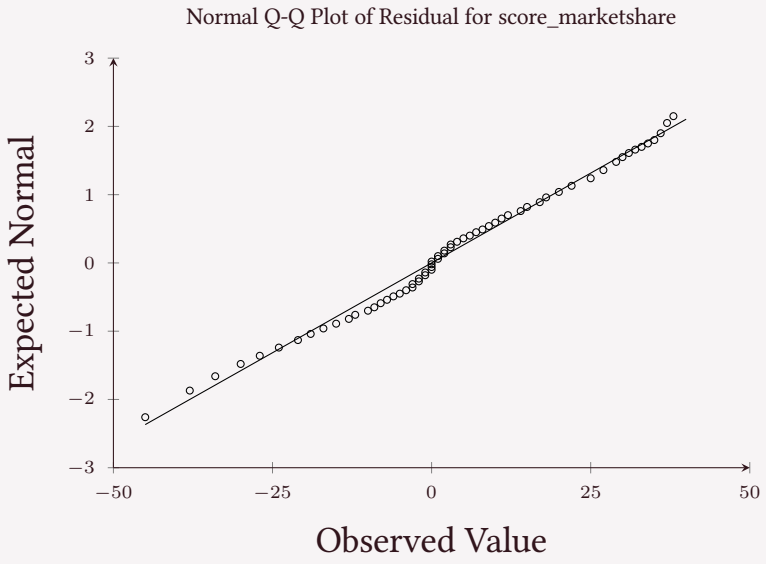
Levene Statistic	df1	df2	Sig.
2.215	2	57	.119

Tests of Normality

	Kolmogorov-Smirnov[a]			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Residual for score_marketshare	.075	60	.200*	.974	60	.235

*. This is a lower bound of the true significance.
a. Lilliefors Significance Correction

We also inspect the Q-Q plot of the residuals:



a. Complete the ANOVA table and test whether that the average market share score differs across the three conditions.

b. Discuss whether the assumptions of the F-test are satisfied.

c. Compute the proportion of the variation in the dependent variable that is due to differences between the three conditions.

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Tip: The Five Steps of ANOVA

ANOVA is a parametric test that compares the means of more than 2 independent samples.

1. **Set up the test:** state the hypotheses and $\alpha = 0.05$:

$$H_0 : \mu_1 = \cdots = \mu_G$$

$$H_A : \text{at least one } \mu_g \text{ is different}$$

2. **Check the assumptions:** in this order:

   (i) Observations are independent.

   (ii) Model errors are Normally distributed (tested with the Kolmogorov-Smirnov test). With a large sample this is not needed, since the CLT ensures the test statistic has an approximate F-distribution.

   (iii) Error variances are equal across groups (tested with Levene's test).

3. **Find summary statistics:** $n_i, \overline{x}_i, s_i^2$ for every group, then N and the size-weighted overall mean $\overline{x}$.

4. **Complete the table:**

	Sum of squares	df	Mean square	F	p
Between groups	$\sum_{i=1}^{G} n_i(\overline{x}_i - \overline{x})^2$	$G-1$	$\frac{SS_{between}}{G-1}$	$\frac{MS_{between}}{MS_{within}}$	$P(F \geq f)$
Within groups	$\sum_{i=1}^{G}(n_i - 1)s_i^2$	$N-G$	$\frac{SS_{within}}{N-G}$		
Total	$\sum_{i=1}^{N}(x_i - \overline{x})^2$	$N-1$			

5. **Decide:** always right-tailed, since only a large F counts as evidence against H_0:

$$\begin{cases} p < \alpha & \rightarrow \text{ reject } H_0 \\ p \geq \alpha & \rightarrow \text{ fail to reject } H_0 \end{cases}$$

Solution 18: a. Complete the ANOVA table and test whether the average score differs across the conditions

Step 1: Set up the test.

H_0 states that the average market share score is the same in all three conditions, while H_A states that at least one condition's mean differs:

$$H_0 : \mu_{control} = \mu_{competition} = \mu_{cooperation}$$

$$H_A : \text{at least one } \mu_g \text{ is different}$$

$$\alpha = 0.05$$

Step 2: Check the assumptions.

The assumptions are discussed in part b.

Step 3: Find summary statistics.

There are $G = 3$ conditions with 20 subjects each (a balanced design), so $N = 60$. The sums of squares and degrees of freedom are already given in the ANOVA table.

Step 4: Compute the mean squares.

Between groups:

$$SS_{between} = 10290.700$$

$$df_{between} = 3 - 1 = 2$$

$$MS_{between} = \frac{10290.700}{2} = 5145.3500$$

Within groups:

$$SS_{within} = 22982.900$$

$$df_{within} = 60 - 3 = 57$$

$$MS_{within} = \frac{22982.900}{57} \approx 403.2088$$

Step 5: Find the test value.

$$F = \frac{5145.3500}{403.2088} \approx 12.7610 \sim \mathcal{F}_{2,\,57}$$

Step 6: Find the p-value and decide.

$$p = P(F \geq 12.7610) \approx 0.0000$$

Since $p < \alpha$ $(0.0000 < 0.05)$, we reject H_0. We conclude that the average market share score differs across the three conditions.

	Sum of squares	df	Mean square	F	p
Between groups	10290.700	2	5145.3500	12.7610	0.0000
Within groups	22982.900	57	403.2088		
total	33273.600	59			

Answer: We conclude with 95% confidence that the average market share score differs across the three conditions.

Solution 18: b. Discuss whether the assumptions of ANOVA are satisfied

ANOVA comes with three assumptions:

1. Observations are independent.
2. Model errors are Normally distributed.
3. Error variances are equal across groups.

Step 1: Check independence.

Each subject is randomly assigned to one of the three conditions and plays the game on their

own, so we can assume that the observations are independent.

Step 2: Check Normality of the model errors.
The Tests of Normality table tests whether the residuals are Normally distributed. In the Kolmogorov-Smirnov columns, we read:

$$Sig. = 0.200$$

Since $Sig. \geq \alpha$, we cannot reject that the residuals are Normally distributed (the Shapiro-Wilk test gives the same result, with $Sig. = 0.235$). The points in the Q-Q plot also lie close to the straight line. This assumption is satisfied.

Step 3: Check equal variances.
The Test of Homogeneity of Variances table gives the test of Levene. In the Sig. column, we read:

$$Sig. = 0.119$$

Since $Sig. \geq \alpha$, we cannot reject that the error variances are equal across the three conditions. This assumption is satisfied.

Answer: All three assumptions of ANOVA are satisfied.

Tip: Effect Size: η^2

η^2 is the proportion of the total variation that is explained by the differences *between* groups:

$$\eta^2 = \frac{SS_{between}}{SS_{total}}$$

It plays the same role as R^2 in regression.

Solution 18: c. Proportion of variation explained

The effect size η^2 is the proportion of the total variation in the market share score that is due to the differences between the three conditions:

$$\eta^2 = \frac{SS_{between}}{SS_{total}} = \frac{10290.700}{33273.600} \approx 0.3093$$

In other words, about a third of why the market share score differs from subject to subject is explained by which condition they were in. The rest is due to other things, not the condition.

Answer: About 30.93% of the variation in the market share score is explained by the differences between the three conditions.

Question 19

In a random sample ($n = 1000$), the sample Pearson correlation equals .07. Test whether the population correlation is positive with 99% reliability. Discuss whether the assumptions of the test are satisfied.

Tip: Pearson Correlation Test

The **Pearson correlation test** is a parametric test used to test whether two quantitative variables have a linear relation in the population.

$$H_0 : \rho = 0 \qquad H_A : \rho \neq 0$$

with ρ the population correlation. It comes with one assumption: (X, Y) has a bivariate Normal distribution (the scatterplot has the shape of an ellipse), or $n > 50$.

Tip: Pearson Correlation Test: Mechanics

In order to do the Pearson correlation test, we use the estimator T. Here are the steps to find it, and its distribution under H_0:

1. Find the sample correlation r and the sample size n.
2. Calculate the estimator:

$$T = R\sqrt{\frac{n-2}{1-R^2}}$$

Under H_0:

$$T \sim t_{n-2}$$

Solution 19: Testing whether the population correlation is positive

Step 1: Define the hypotheses.

We test whether the population correlation is positive. With 99% reliability, the significance level is 0.01:

$$H_0 : \rho \leq 0$$

$$H_A : \rho > 0$$

$$\alpha = 0.01$$

Step 2: State the estimator.

For the Pearson correlation test, we use the estimator T:

$$T = R\sqrt{\frac{n-2}{1-R^2}}$$

Under H_0, T has a t-distribution:

$$T \sim t_{n-2}$$

This is exact if (X, Y) is bivariate Normal, and approximate if $n > 50$ by the CLT.

Step 3: Find the test value.

The sample correlation and the sample size are given:

$$r = 0.07$$

$$n = 1000$$

Plugging these into the estimator:

$$t = 0.07\sqrt{\frac{1000-2}{1-0.07^2}} = 0.07\sqrt{\frac{998}{0.9951}} \approx 2.2168$$

with $df = n - 2 = 998$.

Step 4: Find the p-value.

Since H_A says the correlation is positive, we need the probability of a value at least as large as

t:

$$p = P(T \geq 2.2168) = \texttt{tcdf}(2.2168,\ \texttt{1E99},\ 998) \approx 0.0134$$

Step 5: Make the decision.

The p-value is large, $p > 0.01$, so we cannot reject H_0.

Here we also have to check the assumption of the Pearson correlation test:

- (X, Y) has a bivariate Normal distribution, or $n > 50$.

Here $n = 1000 > 50$, so this assumption is satisfied, even without knowing whether (X, Y) has a bivariate Normal distribution.

Answer: We cannot conclude, with 99% confidence, that the population correlation is positive - it could be 0 or negative.

Question 20

A HR manager wants to evaluate the effect of the number of training days on the performance of sales managers. The manager has sales performance scores for three groups: employees with 1, 2 or 3 days of training.

We run one-way anova with as dependent variable **sales_performance** and with the **number of training days** as factor. Part of the SPSS output is shown below:

Report

sales performance

number of training days	Mean	N	Std. Deviation
1	63.5798	20	13.50858
2	73.5677	20	10.60901
3	79.2792	20	4.40754
Total	72.1422	60	12.00312

Tests of Normality

	Kolmogorov-Smirnov[a]			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Residual for score	.123	60	.025	.957	60	.032

a. Lilliefors Significance Correction

Levene's Test of Equality of Error Variances

Dependent Variable: sales performance

F	df1	df2	Sig.
4.637	2	57	.014

a. Design: Intercept + number_of_trainingdays

Multiple Comparisons

Dependent Variable: sales performance Tukey HSD

(I) number of training days	(J) number of training days	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
1	2	-9.98789*	3.23759	.009	-17.7789	-2.1969
	3	-15.69947*	3.23759	.000	-23.4905	-7.9085
2	1	9.98789*	3.23759	.009	2.1969	17.7789
	3	-5.71158	3.23759	.191	-13.5026	2.0794
3	1	15.69947*	3.23759	.000	7.9085	23.4905
	2	5.71158	3.23759	.191	-2.0794	13.5026

*. The mean difference is significant at the 0.05 level.

a. Use an F-test to investigate whether the number of training days affects the average sales performance of sales managers.

b. Discuss whether the assumptions of the F-test are violated at the 1% significance level.

c. How many training days would you recommend for a future sales training?

Tip: The Five Steps of ANOVA

ANOVA is a parametric test that compares the means of more than 2 independent samples.

1. **Set up the test:** state the hypotheses and $\alpha = 0.05$:

$$H_0 : \mu_1 = \cdots = \mu_G$$

$$H_A : \text{at least one } \mu_g \text{ is different}$$

2. **Check the assumptions:** in this order:

   (i) Observations are independent.

   (ii) Model errors are Normally distributed (tested with the Kolmogorov-Smirnov test). With a large sample this is not needed, since the CLT ensures the test statistic has an approximate F-distribution.

   (iii) Error variances are equal across groups (tested with Levene's test).

3. **Find summary statistics:** $n_i, \overline{x}_i, s_i^2$ for every group, then N and the size-weighted overall mean $\overline{x}$.

4. **Complete the table:**

	Sum of squares	df	Mean square	F	p
Between groups	$\sum_{i=1}^{G} n_i(\overline{x}_i - \overline{x})^2$	$G-1$	$\frac{SS_{between}}{G-1}$	$\frac{MS_{between}}{MS_{within}}$	$P(F \geq f)$
Within groups	$\sum_{i=1}^{G}(n_i-1)s_i^2$	$N-G$	$\frac{SS_{within}}{N-G}$		
Total	$\sum_{i=1}^{N}(x_i - \overline{x})^2$	$N-1$			

5. **Decide:** always right-tailed, since only a large F counts as evidence against H_0:

$$\begin{cases} p < \alpha & \rightarrow \text{ reject } H_0 \\ p \geq \alpha & \rightarrow \text{ fail to reject } H_0 \end{cases}$$

Solution 20: a. Testing whether the number of training days affects the average sales performance

Step 1: Set up the test.

H_0 states that the average sales performance is the same for 1, 2 and 3 training days, while H_A states that at least one group's mean differs:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

$$H_A : \text{at least one } \mu_g \text{ is different}$$

$$\alpha = 0.05$$

Step 2: Check the assumptions.

The assumptions are discussed in part b.

Step 3: Find summary statistics.

The summary statistics for each group are already given in the Report table: $n_i = 20$ for every

group, $N = 60$, and the overall mean is $\overline{x} = 72.1422$.

Step 4: Compute the sums of squares.

Between groups:

$$SS_{between} = 20(63.5798 - 72.1422)^2 + 20(73.5677 - 72.1422)^2 + 20(79.2792 - 72.1422)^2 \approx 2525.6703$$

$$df_{between} = 3 - 1 = 2$$

$$MS_{between} = \frac{2525.6703}{2} \approx 1262.8351$$

Within groups:

$$SS_{within} = 19(13.50858)^2 + 19(10.60901)^2 + 19(4.40754)^2 \approx 5974.7255$$

$$df_{within} = 60 - 3 = 57$$

$$MS_{within} = \frac{5974.7255}{57} \approx 104.8197$$

Step 5: Find the test value.

$$F = \frac{1262.8351}{104.8197} \approx 12.0477 \sim \mathcal{F}_{2,\,57}$$

Step 6: Find the p-value and decide.

$$p = P(F \geq 12.0477) \approx 0.0000$$

Since $p < \alpha$ ($0.0000 < 0.05$), we reject H_0. We conclude that the number of training days affects the average sales performance.

	Sum of squares	df	Mean square	F	p
Between groups	2525.6703	2	1262.8351	12.0477	0.0000
Within groups	5974.7255	57	104.8197		
total	8500.3958	59			

Answer: We conclude with 95% confidence that the number of training days affects the average sales performance of sales managers.

Solution 20: b. Discuss whether the assumptions of ANOVA are violated at the 1% significance level

ANOVA comes with three assumptions:

1. Observations are independent.
2. Model errors are Normally distributed.
3. Error variances are equal across groups.

Here we use $\alpha = 0.01$ for the assumption tests, as the question asks.

Step 1: Check independence.
Nothing in the question suggests that the sales managers influence each other's performance, so we can assume that the observations are independent.

Step 2: Check Normality of the model errors.
The Tests of Normality table tests whether the residuals are Normally distributed. In the Kolmogorov-Smirnov columns, we read:

$$Sig. = 0.025$$

Since $Sig. \geq 0.01$, we cannot reject that the residuals are Normally distributed (the Shapiro-Wilk test gives the same result, with $Sig. = 0.032$). This assumption is not violated.

Step 3: Check equal variances.
Levene's Test of Equality of Error Variances table gives the test of Levene. In the Sig. column, we read:

$$Sig. = 0.014$$

Since $Sig. \geq 0.01$, we cannot reject that the error variances are equal across the three groups. This assumption is not violated.

Answer: At the 1% significance level, none of the assumptions of ANOVA are violated.

Tip: Tukey Post-Hoc Test

Once ANOVA finds that at least one group differs, a **Tukey post-hoc test** tells us specifically which pairs of groups differ. It is really just an independent-samples t-test, repeated for every pair of groups:

$$H_0 : \mu_i = \mu_j \qquad H_A : \mu_i \neq \mu_j$$

The problem is that running many tests back to back increases the chance that at least one of them rejects H_0 just by chance, even if every H_0 is actually true - the more tests we run, the more likely we are to make at least one mistake. Tukey corrects the p-values so that the probability of making at least one such mistake, across the whole set of comparisons, stays at $\alpha = 0.05$.

Solution 20: c. How many training days to recommend

To see which pairs of groups differ, we look at the Tukey post-hoc tests in the Multiple Comparisons table. In the Sig. column, we read:

$$\text{1 vs 2 days}: \ Sig. = 0.009$$

$$\text{1 vs 3 days}: \ Sig. = 0.000$$

$$\text{2 vs 3 days}: \ Sig. = 0.191$$

Managers with 1 day of training have a significantly lower average sales performance than managers with 2 days and with 3 days, since both $Sig. < 0.05$. But the difference between 2 and 3 days is not significant, since $Sig. = 0.191 \geq 0.05$: a third day of training does not significantly improve the average sales performance.

Answer: We would recommend 2 training days: they give a significantly higher average sales performance than 1 day, and a third day does not add a significant improvement.

Chapter 2 - Regression

Chapter 3 - Time Series

Exam: 2022 January

Exam: 2022 August

Exam: 2024 Example 1

Exam: 2024 Example 2

Exam: 2024 January

Exam: 2024 August

Exam: 2025 January

Exam: 2025 August

Exam: 2026 Example

Exam: 2026 January

Exam: Unknown 1

Exam: Unknown 2

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